
Reshaping Reasoning in LLMs: A Theoretical Analysis of RL Training Dynamics through Pattern Selection

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Abstract

1 Reinforcement learning (RL) has become a central mechanism for improving the
2 reasoning capabilities of large language models, yet the training dynamics through
3 which RL changes model behavior remain insufficiently characterized. This paper
4 explains those dynamics by combining fine-grained empirical analysis with a
5 formal model of reasoning-pattern selection. Across RL training, our reasoning-
6 pattern-level and token-level analyses show that individual reasoning patterns
7 retain relatively stable success rates, while RL primarily changes the probability
8 ranks of a sparse subset of critical tokens; these localized token-level changes
9 reshape the distribution over reasoning patterns and thereby determine model
10 performance. Motivated by these observations, we develop a theoretical framework
11 for two representative reward mechanisms: reinforcement learning with verifiable
12 rewards (RLVR) and reinforcement learning with the model’s internal feedback
13 (RLIF). For RLVR, we identify two convergence regimes: one in which the model
14 efficiently concentrates on the optimal reasoning pattern, and another in which
15 optimization is delayed by an entanglement stage, demonstrating the decisive role of
16 the base model’s initial reasoning quality. For RLIF, we show why internal rewards
17 can induce early performance gains while also allowing convergence to lower-
18 performing states under continued training. Extensive experiments and simulations
19 substantiate the empirical and theoretical claims, providing a coordinated account
20 of how RL reshapes reasoning in language-model post-training.

21 1 Introduction

22 Recent reasoning models, including Gemini2.5 [8], Qwen3 [40], and DeepSeek-R1 [14], achieve
23 exceptional performance on complex logical tasks such as mathematics [30, 44, 41] and programming
24 [49, 40, 8]. Reinforcement learning (RL) is a principal post-training mechanism behind these gains,
25 and its empirical success has made RL-based reasoning enhancement a defining technique for
26 contemporary large language models.

27 This success has prompted a growing effort to identify the mechanisms by which RL changes LLM
28 behavior. By comparing pass@k performance, Yue et al. [42] show that RL-post-trained models
29 often struggle to exceed the coverage of their base models, suggesting that RL may not primarily
30 create fundamentally new reasoning patterns. From an entropy perspective, Cui et al. [9], Zhang
31 et al. [45] prove that RL-based methods can reduce policy entropy, and Cui et al. [9] further connects
32 entropy reduction to empirical performance. Complementary analyses by Wang et al. [33], Huan
33 et al. [17] indicate that RL primarily modifies a sparse set of critical tokens. These findings establish
34 important partial views, but they do not yet provide an integrated empirical and theoretical account of
35 the training dynamics that connect token-level changes, reasoning-pattern selection, and downstream
36 accuracy.

We address this gap by conducting a systematic analysis of RL training at both the reasoning-pattern and token levels. Focusing on training dynamics rather than only endpoint performance, we examine next-token ranking shifts and use LLM-based and rule-based procedures to extract, classify, and track reasoning patterns in model responses. This analysis measures both the distribution of patterns selected during training and the success rate associated with each pattern. Relative to prior work [17, 42], our experiments make the mechanism more explicit: RL primarily optimizes a sparse subset of critical tokens, thereby *reshaping reasoning pattern distributions* to change model performance, while the intrinsic success rate of each individual pattern remains comparatively stable. These empirical regularities supply the basis for a mathematical account of RL training.

Building on these findings, we formulate reasoning as a two-stage question-reason-answer process $q \rightarrow r \rightarrow a$: the model first selects a reasoning pattern conditioned on the question, represented by $\pi(r|q)$, and then generates an answer conditioned on both the question and the selected pattern, represented by $\pi(a|r, q)$. Within this framework, we analyze two representative reward types for LLM post-training: verifiable reward (RLVR) [14, 30] and the model’s internal feedback (RLIF) [47, 4]. For RLVR, the analysis proves convergence toward the reasoning pattern r^* with the highest success rate and distinguishes two regimes: strong initial reasoning quality enables rapid concentration on the optimal pattern, whereas weaker initialization can induce an entanglement stage that substantially slows optimization. For RLIF, the framework explains how internal rewards can improve performance early in training while also showing why continued optimization may converge to states that underperform the base model. Additional experiments and simulations confirm these theoretical predictions.

The paper makes three connected contributions:

- We provide a fine-grained empirical account of RL training dynamics, extending prior analyses [17, 42] by combining next-token ranking analysis with LLM-based and rule-based reasoning-pattern analysis. The experiments show that RL primarily modifies a sparse subset of critical tokens, thereby *reshaping reasoning pattern distributions* to affect model performance, while the success rates of individual patterns remain relatively stable throughout training.
- We introduce a formal two-stage framework for reasoning, $q \rightarrow r \rightarrow a$, and use it to analyze the training dynamics of two representative RL-based approaches. For RLVR, we prove convergence to the highest-success reasoning pattern and characterize rapid convergence for strong base models versus entanglement-stage optimization for weaker models; for RLIF, we explain both early performance improvement and possible convergence to worse-than-base performance.
- We substantiate the theoretical analysis with case studies and numerical simulations that reproduce the predicted RLVR regimes and RLIF behavior. Together, the empirical and theoretical results connect observable post-training behavior to a precise mechanism of reasoning-pattern selection.

2 Related Works

Reinforcement Learning for LLMs Reinforcement learning has become a central post-training instrument for large language models (LLMs), with demonstrated impact on preference alignment [25, 48] and on complex mathematical and programming tasks [30, 20, 22]. The reward signal is the organizing component of this paradigm: conventional RL pipelines often rely on reward models trained from human-annotated data [26, 25, 2], whereas recent reasoning-oriented systems increasingly exploit rewards that are easier to obtain or verify. This shift is most visible in reinforcement learning with verifiable rewards (RLVR) [30, 40, 14] and reinforcement learning with internal feedback (RLIF) [47, 4]. In parallel, the empirical success of these methods has motivated algorithmic variants such as GRPO [30], DAPO [41], and VAPO [43], making the mechanisms of RL-based reasoning improvement an increasingly important object of study.

Theoretical Analyses and Mechanism Interpretation of RL for LLMs. The rapid adoption of RL for LLM post-training has also generated a substantial literature on theoretical explanation and mechanism interpretation. Early analyses primarily addressed higher-level questions, including reward design [28, 18, 38] and the development of improved training algorithms [37, 11, 21]. As the field has moved toward simpler reward structures such as RLVR and RLIF, the central question has become why these signals can improve reasoning at all. Shao et al. [29] argue that spurious rewards can unlock latent skills acquired during pretraining, whereas Agarwal et al. [4] explain

90 performance gains through entropy minimization. Gandhi et al. [13] identify cognitive behaviors,
 91 including backtracking, that correlate with improved reasoning. Additional evidence from [46] shows
 92 that RL fine-tuning can amplify pretrained behaviors and drive convergence toward dominant output
 93 formats, a phenomenon further examined by Wang et al. [34], Cui et al. [10] through entropy collapse.
 94 Our work advances this line of inquiry by placing reasoning-pattern selection at the center of the
 95 analysis: it connects observable training dynamics to a formal framework that explains how different
 96 reward mechanisms reshape the distribution over reasoning patterns.

97 3 Preliminaries

98 **Reinforcement Learning for LLMs** Let π_θ denote a language model with parameters θ , interpreted
 99 as the policy optimized during post-training. For an input question $\mathbf{x} = (x_0, x_1, \dots, x_n)$, the policy
 100 samples an answer sequence $\mathbf{y} = (y_0, y_1, \dots, y_m)$. The reinforcement-learning objective balances
 101 expected reward against a KL penalty that constrains the learned policy relative to a reference model:

$$\phi_{\text{RL}}(\theta) = \mathbb{E}_{\mathbf{x} \sim \mathcal{D}, \mathbf{y} \sim \pi_\theta(\cdot | \mathbf{x})} [r_\phi(\mathbf{x}, \mathbf{y})] - \beta \mathbb{D}_{\text{KL}} [\pi_\theta(\mathbf{y} | \mathbf{x}) \parallel \pi_{\text{ref}}(\mathbf{y} | \mathbf{x})], \quad (3.1)$$

102 where π_{ref} is the base reference policy, and β controls the strength of the KL regularizer that limits
 103 excessive deviation between π_θ and π_{ref} .

104 The reward function $r_\phi(\mathbf{x}, \mathbf{y})$ may be instantiated as a trained reward model [25] or as a rule-based
 105 scoring function [30]. This paper focuses on two representative reward mechanisms that are central to
 106 recent reasoning post-training: verifiable reward (RLVR) [14, 30] and the model’s internal feedback
 107 (RLIF) [47, 4].

108 In RLVR, $r_\phi(\mathbf{x}, \mathbf{y})$ directly evaluates whether the generated answer $\mathbf{y}$ matches the ground-truth
 109 answer for question $\mathbf{x}$. A typical binary verifier is defined as:

$$r_\phi(\mathbf{x}, \mathbf{y}) = \begin{cases} 1 & \text{if } \mathbf{y} = \text{the ground truth of } \mathbf{x}, \\ 0 & \text{otherwise.} \end{cases} \quad (3.2)$$

110 Unlike RLVR, RLIF [47, 4] constructs rewards entirely from intrinsic, model-derived signals rather
 111 than external correctness labels. Following Zhao et al. [47], we define the RLIF reward as the negative
 112 average KL divergence between a uniform distribution U over the vocabulary $\mathcal{V}$ and the model’s
 113 next-token distribution:

$$r_\phi(\mathbf{x}, \mathbf{y}) = \frac{1}{|\mathbf{y}|} \sum_{i=1}^{|\mathbf{y}|} \mathbb{D}_{\text{KL}}(U \parallel \pi_\theta(\cdot | \mathbf{x}, \mathbf{y}_{<i})) = -\frac{1}{|\mathbf{y}| \cdot |\mathcal{V}|} \sum_{i=1}^{|\mathbf{y}|} \sum_{j=1}^{|\mathcal{V}|} \log(|\mathcal{V}| \cdot \pi_\theta(\cdot | \mathbf{x}, \mathbf{y}_{<i})). \quad (3.3)$$

114 The remainder of the paper uses these two objectives as controlled cases for analyzing how reward
 115 structure governs reasoning-pattern selection and, consequently, model performance during RL
 116 training.

117 4 Experimental Exploration for RL Training

118 To characterize the RL training process, we conduct systematic experiments across reward mech-
 119 anisms and analyze the resulting dynamics at progressively finer resolutions. We begin with a
 120 controlled overview of training trajectories, which shows that RLVR produces steady performance
 121 gains, whereas RLIF can be unstable and may ultimately degrade performance. We then examine
 122 responses at the reasoning-pattern level, where RLVR improves accuracy by increasing the prevalence
 123 of higher-success patterns, while RLIF does not consistently specialize toward those patterns. Finally,
 124 token-level analysis identifies the local mechanism behind these distributional shifts: across identical
 125 responses, RL changes the probability ranks of only a small fraction of tokens. Compared with
 126 previous works [17, 42], these experiments provide direct evidence that RL primarily optimizes a
 127 sparse subset of critical tokens and thereby *reshapes reasoning pattern distributions* to affect model
 128 performance.

129 4.1 Training Procedure Overview for RLVR and RLIF

130 We first compare the complete training procedures of RLVR and RLIF under matched experimental
 131 conditions. We use Qwen2.5-3B [39] as the base model, train on the MATH dataset [16], and keep
 132 all other settings identical across the two algorithms.

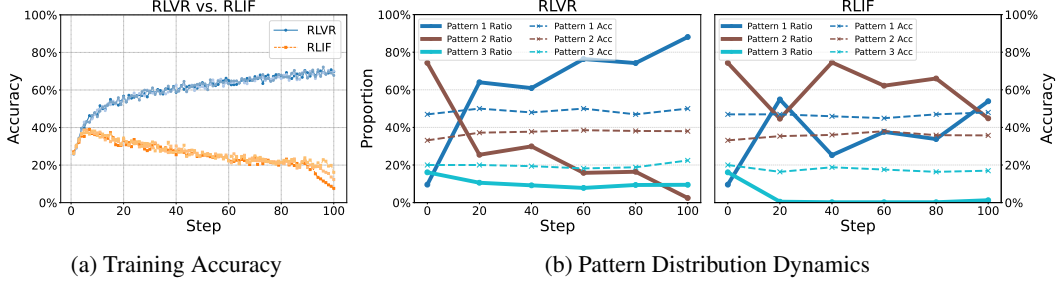


Figure 1: **(a)** Training procedures for RLVR and RLIF on the MATH dataset. Across multiple random-seed runs, RLVR provides stable gains, whereas RLIF initially improves performance and then declines. **(b)** Reasoning-pattern-level comparison. Solid lines report the response proportion of each pattern among all patterns (left vertical axis), and dotted lines report the accuracy associated with that pattern (right vertical axis). During RLVR, the model increasingly selects higher-accuracy patterns and reduces lower-accuracy patterns, while RLIF exhibits unstable pattern dynamics. For readability, patterns are ordered from high to low by their corresponding accuracy.

Figure 1a shows that the two reward mechanisms induce sharply different trajectories. RLVR yields a stable optimization process in which performance improves continuously and gradually converges. RLIF, by contrast, can improve performance during the early stages but then decline under continued training, in some runs falling below the original base-model performance. This divergence motivates the subsequent fine-grained analysis of how each reward mechanism changes reasoning behavior.

4.2 Reasoning-pattern Level Analysis

To explain the performance disparity between RLVR and RLIF, we next analyze model outputs at the reasoning-pattern level. We collect responses from the base model and use GPT-4o [19] to group them into categories defined by keywords and logical structure, with a description generated for each pattern. These classifications allow us to track both the distribution of reasoning patterns and the success rate associated with each pattern throughout RL training.

Specifically, we analyze how reasoning-pattern frequencies and pattern-specific accuracies evolve under RLVR and RLIF. Figure 1b reports the analysis for Number Theory tasks from the MATH dataset [16]; Appendix A provides additional results across models and datasets. The observed dynamics yield three findings:

- RLVR-trained models consistently shift toward reasoning patterns with higher success rates, explaining the steady improvement in overall accuracy under RLVR.
- RLIF produces unstable reasoning-pattern distributions and does not reliably specialize in the more effective patterns.
- For both methods, the success rate of each individual reasoning pattern remains comparatively stable throughout training.

These results identify reasoning-pattern selection as the immediate source of the performance difference between RLVR and RLIF. We therefore examine token-level rank changes to determine how RL training induces the observed pattern-level redistribution.

4.3 Token-level Analysis

To identify the mechanism behind the reasoning-pattern dynamics, we conduct a token-level analysis inspired by Huan et al. [17], Yue et al. [42]. We first sample responses from the base model on GSM8K [6], MATH [16], and AIME [7]. For these identical responses, we then compare the corresponding token ranks under the base model and the RL-enhanced model, where a token’s

Table 1: Token-ranking change ratio after RL.

Task	Method	Step20	Step40	Step60	Step80	Step100
GSM8K	RLVR	5.2%	6.0%	6.6%	7.1%	7.3%
	RLIF	6.8%	8.0%	8.9%	9.5%	10.1%
MATH	RLVR	5.3%	5.8%	6.1%	6.5%	6.6%
	RLIF	6.1%	7.0%	7.7%	8.3%	8.8%
AIME24	RLVR	5.1%	5.7%	5.9%	6.1%	6.3%
	RLIF	5.6%	6.5%	7.4%	8.2%	8.6%

rank is the position of its probability among all vocabulary tokens conditioned on the preceding context. Table 1 shows that individual-token ranks remain largely stable: in most settings, fewer than 10% of tokens experience a rank shift. Thus, RL selectively changes the probabilities of a sparse set of critical tokens while leaving most of the generated reasoning trajectory unchanged.

Together, these experiments establish a coherent empirical picture of RL training dynamics: *RL primarily optimizes a sparse subset of critical tokens, thereby reshaping reasoning pattern distributions to affect model performance. Moreover, the intrinsic success rate of individual patterns remains relatively stable during training.*

These empirical findings motivate the theoretical framework developed next, in which RL improves or degrades performance by changing the distribution over reasoning patterns rather than by uniformly improving every pattern.

5 Theoretical Characterization of RL Training Dynamics

The empirical results indicate that RL changes reasoning primarily by redistributing probability mass across reasoning patterns. We now formalize this mechanism and analyze the training dynamics it implies. The framework abstracts reasoning as a two-stage process: the model first selects a reasoning pattern and then deduces an answer conditioned on that pattern. Under the empirical observation that pattern-specific success rates are comparatively stable, RL mainly optimizes the first stage. This formulation yields distinct explanations for the two reward mechanisms studied above: for RLVR, we characterize two regimes with different convergence behavior, and for RLIF, we explain why internal rewards can produce early gains yet eventually converge to states that underperform the base model.

5.1 A Theoretical Framework for Reasoning Models

Abstract Reasoning Process. We formalize reasoning as a structured stochastic process. Given a question q , the model first selects a reasoning pattern r from the candidate set $\mathcal{R} = \{r_1, r_2, \dots\}$ and then generates a final answer $a \in \mathcal{A} = \{a_1, a_2, \dots\}$ conditioned on that pattern. Pattern r_i is selected with probability $p(r_i|q)$, and different patterns may have different success rates. Within this abstraction, the RLIF optimization objective (Eq. 3.3) can be reformulated as follows¹:

$$r_\phi(\{r, a\}, q) := - \left(\frac{1}{|\mathcal{R}|} \sum_{r_i \in \mathcal{R}} \log(|\mathcal{R}| \cdot \pi_\theta(r_i|q)) + \frac{1}{|\mathcal{A}|} \sum_{a_j \in \mathcal{A}} \log(|\mathcal{A}| \cdot \pi_\theta(a_j|q, r)) \right) \quad (5.1)$$

Policy Parameterization. Let $\mathcal{V} = \{q\} \cup \mathcal{R} \cup \mathcal{A}$ denote the vocabulary set and $\mathcal{Y} = (r, a)_{r \in \mathcal{R}, a \in \mathcal{A}}$ denote the set of output sequences. Given question q , the language model defines an autoregressive distribution over output sequences $y \in \mathcal{Y}$:

$$(\text{general policy}) \quad \pi_\theta(y|x) = \prod_{l=1}^{|y|} \pi_\theta(y_l|x, y_{<l}) = \prod_{l=1}^{|y|} \text{softmax}(f_\theta(x, y_{<l}))_{y_l}, \quad (5.2)$$

where $f_\theta : \mathcal{V} \rightarrow \mathbb{R}^{|\mathcal{V}|}$ is parameterized by θ , and the l -th token is predicted from the preceding context $y_{<l}$. The next-token distribution is $\text{softmax}(z)_v := \exp(z_v) / \sum_{v' \in \mathcal{V}} \exp(z_{v'})$ for $z = f_\theta(x, y_{<l})$.

Because practical language models are highly complex, deriving optimization guarantees that directly describe RL training dynamics is difficult [3, 23]. Following previous work [27, 24, 10, 45], we therefore analyze a *tabular policy* parameterization. This parameterization is a special case of Eq. 5.2 in which each output is assigned a trainable logit for the corresponding last token, i.e., for $\theta \in \mathbb{R}^{|\mathcal{V}| \times |\mathcal{V}|}$:

$$(\text{tabular policy}) \quad \pi_\theta(y_l|x, y_{<l}) = \pi_\theta(y_l|y_{l-1}) = \text{softmax}(\theta_{:,y_{l-1}})_{y_l}, \quad (5.3)$$

where $\theta_{:,y_{l-1}} \in \mathbb{R}^{|\mathcal{V}|}$ is the column of θ corresponding to y_{l-1} .

¹For simplicity, we omit the normalizing coefficient $1/|y|$ from Eq. 3.3, which can be treated as a constant $1/2$ in our framework.

203 **Optimization for RL.** To analyze training dynamics, we combine the tabular policy in Eq.5.3 with
 204 the RL objective $\phi_{\text{RL}}(\theta)$ in Eq.3.1. We study the policy-gradient dynamics in the small-learning-rate
 205 limit via gradient flow:

$$\frac{d}{dt}\theta(t) = \nabla\phi_{\text{RL}}(\theta(t)), \quad t \geq 0 \quad (5.4)$$

206 where $\theta(t)$ denotes the parameters of policy $\pi_{\theta(t)}$ at training time t , initialized at $\theta(0) = \theta_{\text{ref}}$.

207 The experiments show that the success rates of individual reasoning patterns often remain stable
 208 during training. We model this regularity by separating pattern selection from answer deduction:
 209 optimizing the mapping from questions to reasoning patterns is treated as substantially easier than
 210 optimizing the conditional mapping from a selected pattern to the final answer. This motivates the
 211 following assumption:

212 **Assumption 5.1.** *The success rate for each reasoning pattern $\mathbf{r}_i \in \mathcal{R}$ to provide the correct answer*
 213 *$\mathbf{r}^*$, defined as $\pi_{\theta}(\mathbf{a}^*|\mathbf{q}, \mathbf{r}_i)$, remains constant during training.*

214 In the remainder of this section, we adopt Assumption 5.1 and write $p^*(\mathbf{r}) = \pi_{\theta}(\mathbf{a}^*|\mathbf{q}, \mathbf{r})$ for the
 215 fixed success rate of pattern $\mathbf{r}$. We first derive the optimal policy for the RL objective (Eq.3.1) under
 216 the general autoregressive policy (Eq.5.2); we then analyze RLVR and RLIF training dynamics under
 217 the tabular policy (Eq. 5.3).

218 5.2 The Optimal Policy for RL

219 The empirical results show that RLVR steadily improves model performance by favoring higher-
 220 success reasoning patterns, whereas RLIF produces less stable gains. We first characterize the optimal
 221 policies induced by the RLVR and RLIF objectives:

222 **Proposition 5.2.** *Suppose we maximize the RL objective (Eq 3.1) using a general autoregressive*
 223 *policy (Eq 5.2) and Assumption 5.1 holds. Then, the optimal policy satisfies:*

$$\pi_{\text{opt}}(\mathbf{r}|\mathbf{q}) = \frac{1}{Z} \exp\left(\frac{1}{\beta}R(\mathbf{r})\right) \pi_{\theta_{\text{ref}}}(\mathbf{r}|\mathbf{q}) \text{ for all } \mathbf{r} \in \mathcal{R}, \quad (5.5)$$

224 where $Z = \sum_{\mathbf{r} \in \mathcal{R}} \exp\left(\frac{1}{\beta}R(\mathbf{r})\right) \pi_{\theta_{\text{ref}}}(\mathbf{r}|\mathbf{q})$ is the normalizing coefficient, and $R(\mathbf{r})$ denotes the
 225 reasoning path reward. Specifically:

- 226 • For RLVR, $R(\mathbf{r})$ equals the success rate of reasoning pattern $\mathbf{r}$, i.e., $R_{\text{RLVR}}(\mathbf{r}) = p^*(\mathbf{r})$.
- 227 • for RLIF, $R(\mathbf{r})$ equals to the confidence for the reasoning pattern $\mathbf{r}$ for final answer distribution,
 228 i.e., $R_{\text{RLIF}}(\mathbf{r}) = -\frac{1}{|\mathcal{A}|} \sum_{\mathbf{a} \in \mathcal{A}} \log(\pi_{\theta}(\mathbf{a}|\mathbf{q}, \mathbf{r}))$.

229 Proposition 5.2 characterizes the optimal solutions for RLVR and RLIF. Assuming π_{opt} lies in the
 230 general autoregressive policy class (which includes both LLMs and the tabular policies analyzed
 231 below), the RLVR-optimized model increases the probability of a reasoning pattern $\mathbf{r}$ when its
 232 probability of deducing the correct answer $p^*(\mathbf{r})$ is sufficiently high, i.e., when $\exp(\frac{1}{\beta}p^*(\mathbf{r}))/Z > 1$.

233 RLIF behaves differently: $R_{\text{RLIF}}(\mathbf{r})$ depends on the confidence of the final-answer distribution and
 234 therefore does not distinguish the correct answer $\mathbf{a}^*$ from incorrect alternatives. Thus, the RLVR
 235 optimum is aligned with higher-success reasoning patterns, whereas the RLIF optimum does not by
 236 itself guarantee improved accuracy over the base model.

237 Moreover, because β is typically small in practice (e.g., 0.001) [26, 5, 44], the limiting behavior as
 238 $\beta \rightarrow 0$ is:

$$\pi_{\text{opt}}(\mathbf{r}|\mathbf{q}) = \lim_{\beta \rightarrow 0} \frac{1}{Z} \exp\left(\frac{1}{\beta}R(\mathbf{r})\right) \pi_{\theta_{\text{ref}}}(\mathbf{r}|\mathbf{q}) = \begin{cases} 1 & \mathbf{r} = \arg \max_{\mathbf{r}} R(\mathbf{r}) \\ 0 & \text{otherwise} \end{cases}, \quad (5.6)$$

239 In this limit, the policy becomes deterministic and selects the reasoning pattern $\mathbf{r}$ with the largest $R(\mathbf{r})$,
 240 independent of the initial reference distribution over patterns. The optimal-policy characterization
 241 explains the endpoint preference but not the path taken during training. Because the optimization
 242 landscape is non-convex, it remains necessary to analyze how RLVR reaches high-reward patterns
 243 and why RLIF can nevertheless improve performance at the beginning of training. We therefore study
 244 both methods under the tabular policy in Eq. 5.3.

245 5.3 Training Dynamic Analysis for RLVR

246 The experiments also indicate that the effectiveness of RLVR depends strongly on the initial quality of
 247 the base model. We focus on the probability assigned to the *optimal reasoning pattern*, namely the pat-
 248 tern $\mathbf{r}^*$ with the highest success rate for reaching the correct answer $\mathbf{a}^*$, where $\mathbf{r}^* = \arg \max_{\mathbf{r}} p^*(\mathbf{r})$.
 249 Gradient-flow analysis reveals two regimes. In the first, the probability of the optimal pattern $\pi_{\theta}(\mathbf{r}^*|\mathbf{q})$
 250 increases steadily until it converges to 1. In the second, the model first enters an entanglement stage
 251 in which a suboptimal pattern $\mathbf{r}' \neq \mathbf{r}^*$ slows optimization toward the optimal pattern; after this stage,
 252 the dynamics transition to the first regime and converge to $\mathbf{r}^*$.

253 **Theorem 5.3** (Regime 1: Sufficient Condition for Efficient Convergence). *Consider the RLVR (Eq*
 254 *3.2) with $\beta = 0$ for optimizing objective Eq 3.1, using a tabular policy (Eq 5.3) with Assumption 5.1*
 255 *holds. Let $\mathbf{r}^*$ be the optimal reasoning pattern, if the overall accuracy of the initialized model $\pi_{\theta_{\text{ref}}}$,*
 256 *defined as $\text{ACC}_{\theta_{\text{ref}}} = \sum_{\mathbf{r} \in \mathcal{R}} \pi_{\theta_{\text{ref}}}(\mathbf{r}|\mathbf{q})p^*(\mathbf{r})$, satisfies:*

$$(Regime 1) \quad \text{ACC}_{\theta_{\text{ref}}} > p^*(\mathbf{r}) \text{ for all } \mathbf{r} \in \mathcal{R}, \mathbf{r} \neq \mathbf{r}^*, \quad (5.7)$$

257 *then for any $\epsilon > 0$, there exists $T_1 = \mathcal{O}(\frac{1}{\epsilon})$ such that for $t > T_1$, we have $1 - \pi_{\theta(t)}(\mathbf{r}^*|\mathbf{q}) < \epsilon$.*

258 Theorem 5.3 covers the favorable regime in which the base reference model is sufficiently strong: its
 259 overall accuracy exceeds the success rate of every non-optimal reasoning pattern (Eq 5.7). Under
 260 this condition, RLVR efficiently drives selection toward the optimal reasoning pattern at rate $\mathcal{O}(1/\epsilon)$,
 261 yielding overall accuracy close to the success rate of that optimal pattern.

262 In practice, however, RLVR optimization can be difficult in some settings [44, 36], especially when
 263 the base reference model is weaker. We therefore analyze a second regime in which the model initially
 264 experiences an entanglement stage: the *suboptimal reasoning pattern with the second-highest success*
 265 *rate*, defined as $\mathbf{r}' = \arg \max_{\mathbf{r}, \mathbf{r} \neq \mathbf{r}^*} p^*(\mathbf{r})$, delays optimization toward the optimal pattern:

266 **Theorem 5.4** (Regime 2: Slow Convergence for optimal reasoning pattern). *Consider the RLVR (Eq*
 267 *3.2) with $\beta = 0$ for optimizing objective Eq 3.1, using a tabular policy (Eq 5.3) with Assumption 5.1*
 268 *holds. Let $\mathbf{r}^*$ and $\mathbf{r}'$ be the optimal and second optimal reasoning patterns, if the overall accuracy of*
 269 *the initialized model satisfies:*

$$(Regime 2) \quad p^*(\mathbf{r}') > \text{ACC}_{\theta_{\text{ref}}} > p^*(\mathbf{r}) \text{ for all } \mathbf{r} \in \mathcal{R}/\{\mathbf{r}^*, \mathbf{r}'\}, \quad (5.8)$$

270 *then there exists:*

$$T_0 = \frac{1}{2 - 2\pi_{\theta_{\text{ref}}}(\mathbf{r}'|\mathbf{q})} \left((C_1 \cdot \gamma_{\pi_{\text{ref}}})^{2C_2 \cdot \gamma_{\pi_{\text{ref}}}} - 1 \right), \text{ where } \gamma_{\pi_{\text{ref}}} := \sum_{\mathbf{r} \in \mathcal{R}/\{\mathbf{r}'\}} \frac{\pi_{\theta_{\text{ref}}}(\mathbf{r}|\mathbf{q})}{\pi_{\theta_{\text{ref}}}(\mathbf{r}^*|\mathbf{q})} \quad (5.9)$$

271 *with constants C_1, C_2 depending on the success rates of reasoning patterns, such that for we can*
 272 *guarantee the model transform from regime 2 (Eq 5.8) to regime 1 (Eq 5.7), i.e., $\text{ACC}_{\theta(t)} >$*
 273 *$p^*(\mathbf{r}), \forall \mathbf{r} \in \mathcal{R}, \mathbf{r} \neq \mathbf{r}^*$ for $t \geq T_0$.*

274 Theorem 5.4 analyzes a special regime in which only the optimal and second-optimal reasoning
 275 patterns have success rates above the model’s overall accuracy. This restriction makes the dynamics
 276 tractable while preserving the central difficulty: more general settings in which *at least* two patterns
 277 exceed the average should exhibit analogous behavior, although their analysis would require additional
 278 technical complexity.

279 The central implication of Theorem 5.4 is that RLVR may require T_0 steps before the overall success
 280 rate exceeds the success rate of the suboptimal pattern, thereby entering the favorable regime of
 281 Theorem 5.3 in which the optimal pattern has sufficiently large selection probability. Whereas
 282 Theorem 5.3 gives a convergence time T_1 that is polynomial in $1/\epsilon$, the transition time T_0 in Theorem
 283 5.4 can grow super-exponentially in $\gamma_{\pi_{\text{ref}}}$ —the ratio between the total success rate through $\mathcal{R}/\{\mathbf{r}'\}$
 284 and the success rate through $\mathbf{r}^*$. When the base model assigns very little probability to the optimal
 285 reasoning pattern, $\gamma_{\pi_{\text{ref}}}$ can be large, making T_0 prohibitively long. This prolonged transition is
 286 precisely the entanglement stage.

287 Section 6 provides case studies that illustrate both regimes.

288 5.4 Theoretical Explanation for RLIF

289 Proposition 5.2 shows that RLIF, because it only considers the confidence of the final-answer
 290 distribution, does not distinguish the correct answer $\mathbf{a}^*$ from incorrect alternatives. This explains why

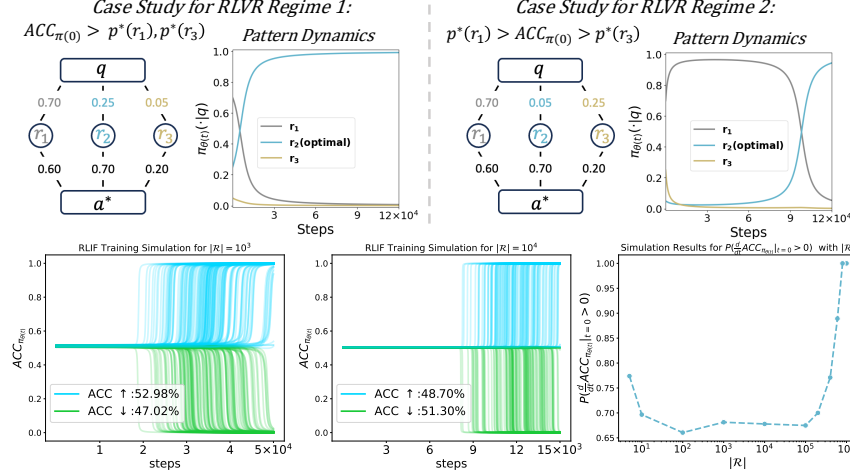


Figure 2: Case studies for RLVR (upper) and training simulations for RLIF (bottom). The upper panels illustrate the two RLVR regimes: rapid convergence (left) and an entanglement phase (right). The bottom panels report, across varying $|\mathcal{R}|$, the probabilities of convergence to different RLIF states (left) and the probability of initial RLIF performance gains (right).

RLIF can ultimately degrade performance relative to the base model, consistent with our empirical observations. At the same time, prior studies report that RLIF can improve model performance without external rewards Agarwal et al. [4], Zhao et al. [47]. We now explain why such improvement can occur during the initial stage of training for a well-trained LLM.

We begin with a well-trained LLM satisfying the following assumption:

Assumption 5.5. *For the base model, the correct answer $\mathbf{a}^*$ has the highest probability across all possible answers, i.e., $\arg \max_{\mathbf{a} \in \mathcal{A}} \sum_{\mathbf{r} \in \mathcal{R}} \pi_{\text{base}}(\mathbf{a}|\mathbf{q}, \mathbf{r}) \pi_{\text{base}}(\mathbf{r}|\mathbf{q}) = \mathbf{a}^*$.*

Because majority voting [35] is an effective way to improve model performance, a well-trained LLM often assigns higher aggregate probability to the correct answer $\mathbf{a}^*$ than to any incorrect answer; Assumption 5.5 therefore captures a plausible property of modern LLMs. Under this condition, we analyze the case $|\mathcal{A}| = 2$, with uniformly distributed reasoning-path success rates and high-entropy reasoning-pattern selection². The accuracy dynamics at $t = 0$ then satisfy the following theorem:

Theorem 5.6 (RLIF Increases Overall Accuracy at Initialization). *Consider the RLIF (Eq 5.1) with $\beta = 0$ for optimizing objective Eq 3.1, using a tabular policy (Eq 5.3) with Assumption 5.1 and 5.5 hold, $\pi_{\theta_{\text{base}}}(\mathbf{r}|\mathbf{q}) = \frac{1}{|\mathcal{R}|}$ for all $\mathbf{r} \in \mathcal{R}$, $|\mathcal{A}| = 2$ and success rate for each reasoning path follows a uniform distribution $p^*(\mathbf{r}) \sim U[0, 1]$. Then when $|\mathcal{R}| \rightarrow +\infty$, the following holds:*

1. The accuracy derivative at $t = 0$ is positive with probability 1, i.e. $P(\frac{d}{dt} ACC_{\theta(t)}|_{t=0} > 0) = 1$.
2. With probability $p = 0.5$, that $\arg \max_{\mathbf{r}} R_{\text{RLIF}}(\mathbf{r}) = \arg \min_{\mathbf{r}} p^*(\mathbf{r})$.

The first result in Theorem 5.6 shows that overall accuracy increases at initialization, providing a theoretical explanation for RLIF’s early performance gains. As training proceeds, however, Proposition 5.2 implies that when $\beta = 0$, the policy converges deterministically to the reasoning pattern $\mathbf{r}$ that maximizes $R_{\text{RLIF}}(\mathbf{r})$. The second result in Theorem 5.6 shows that, with probability 0.5, this maximizing pattern is also the pattern with minimum success rate: $\arg \max_{\mathbf{r}} R_{\text{RLIF}}(\mathbf{r}) = \arg \min_{\mathbf{r}} p^*(\mathbf{r})$. Thus, in this setting, there is a 50% chance that the model converges to the least accurate reasoning path. RLIF can therefore improve performance initially while still converging, under continued training, to states that favor very low-accuracy reasoning paths and may perform worse than the base model.

6 Case Studies and Numerical Simulations

We further substantiate the theoretical analysis through case studies and numerical simulations. For RLVR, Fig. 2 (upper) illustrates the two training regimes characterized in Theorems 5.3 and 5.4. For RLIF, Fig. 2 (bottom) reports repeated simulations in which each reasoning path’s success rate follows a uniform distribution, validating the predicted behavior at both initialization and convergence.

6.1 Case Studies for RLVR

Section 5.3 identifies two regimes in RLVR training dynamics. In the first regime (Eq 5.7), the probability of selecting the optimal reasoning pattern $\pi_\theta(\mathbf{r}^*|\mathbf{q})$ increases monotonically until convergence. In the second regime (Eq 5.8), the model undergoes an initial entanglement phase and requires a substantially longer transition before entering the first-regime dynamics and converging to $\mathbf{r}^*$. We examine both regimes empirically:

- Fig. 2 (upper left) shows that when the reference model satisfies Eq 5.7—that is, when its overall accuracy exceeds the success rates of all non-optimal reasoning patterns—the model rapidly converges to the optimal reasoning pattern.
- The second case corresponds to Theorem 5.4. When a suboptimal reasoning pattern $\mathbf{r}'$ has a success rate above the initial overall accuracy, the model first enters an entanglement phase. The transition period T_0 defined in Theorem 5.4 must pass before rapid optimization begins. Fig. 2 (upper right) illustrates a setting with large $\gamma_{\pi_{\text{ref}}}$ (defined in Eq 5.9), where $\gamma_{\pi_{\text{ref}}} = 6$ because $(0.25 + 0.05)/0.05 = 6$. In this case, T_0 , which scales with $\gamma_{\pi_{\text{ref}}}^{\gamma_{\pi_{\text{ref}}}}$, becomes prohibitively long and substantially delays convergence to the optimal reasoning pattern. The pattern dynamics in Fig. 2 clearly exhibit this entanglement stage, matching the theoretical prediction.

6.2 Numerical Simulations For RLIF

Section 5.4 shows that, under Assumption 5.5 and the base-model distribution specified in Theorem 5.6, the accuracy derivative at $t = 0$ is positive with probability 1 as $|\mathcal{R}| \rightarrow \infty$, while the model has a 50% probability of converging to the least accurate reasoning path. The simulations validate both claims. Fig. 2 (bottom left) reports RLIF training simulations across varying $|\mathcal{R}|$ and shows that the probability of convergence to an improved state ($ACC \uparrow$) is approximately equal to the probability of convergence to a degraded state ($ACC \downarrow$), supporting the second result of Theorem 5.6. Fig. 2 (bottom right) examines the initial training step and shows that, for large $|\mathcal{R}|$, the probability of initial improvement approaches 1, supporting the first result of Theorem 5.6.

7 Conclusions and Limitations

This work provides an empirical and theoretical account of how reinforcement learning reshapes reasoning in LLM post-training. The central conclusion is that RL primarily changes model performance by altering the distribution over reasoning patterns, driven by probability-rank changes in a sparse subset of critical tokens, while the success rates of individual patterns remain comparatively stable. We formalize this mechanism for two representative reward structures, RLVR and RLIF, and validate the resulting predictions through case studies and simulations. The analysis also clarifies the role of base-model quality: RLVR can converge efficiently when the base model already assigns sufficient probability to high-success patterns, but weaker initializations may experience a prolonged entanglement stage; RLIF can improve performance early yet converge to lower-performing states under continued optimization. Several limitations remain important for future work: LLM-identified reasoning patterns require additional interpretability validation, the theoretical abstraction should be extended to more complex real-world reasoning settings, and the assumptions on base-model behavior should be generalized beyond the cases analyzed here.

²Previous work [33] reveals that RL primarily optimizes high-entropy tokens. Here, we consider the highest entropy case: $\pi_{\theta_{\text{base}}}(\mathbf{r}|\mathbf{q}) = 1/|\mathcal{R}|$ for all $\mathbf{r} \in \mathcal{R}$.

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A Additional Experiments

A.1 Pattern Distribution Dynamics on RL-enhanced Qwen2.5-3B

Section 4.2 analyzes reasoning-pattern dynamics on Number Theory tasks sampled from MATH. We extend the same dynamic analysis to Geometry tasks in Figure 3. The pattern trajectories reproduce the main empirical finding: under RLVR, the model increasingly selects higher-accuracy reasoning patterns, whereas RLIF produces unstable pattern selection. These results provide additional support for the empirical and theoretical account developed in the main paper.

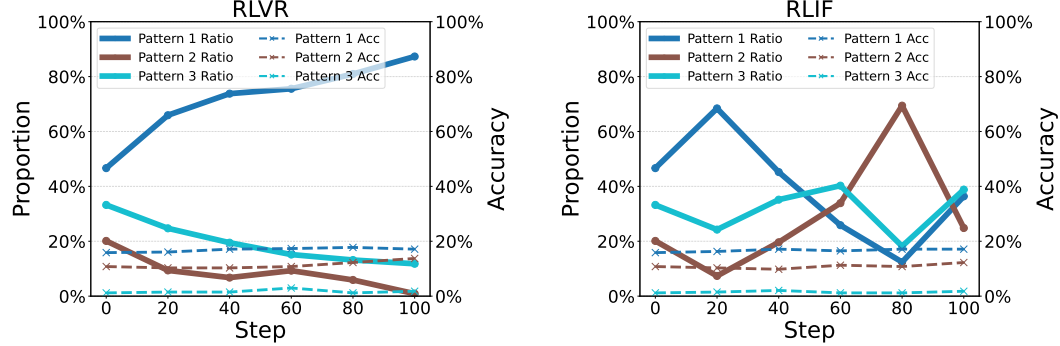


Figure 3: Dynamics of pattern distributions and corresponding accuracies for RL-enhanced models on Geometry tasks. Under RLVR, the model increasingly selects the highest-accuracy pattern (Pattern 1), whereas RLIF exhibits unstable pattern-selection behavior.

A.2 Qwen-2.5-7B-Instruct vs. Qwen-2.5-7B-SimpleRL-Zoo

To assess whether the findings extend across model instances and task categories already covered by our evaluation, we conduct supplementary experiments summarized in Figure 4. We use Qwen-2.5-7B-Instruct [39] as the base model and compare it with Qwen-2.5-7B-SimpleRL-Zoo [44], an open-source RLVR-enhanced model for mathematical reasoning. The evaluation covers number theory, geometry, algebra, calculus, and counting and probability problems sampled from MATH [16], and further includes a problem from AMC23 [1]. Across these settings, RLVR enhancement increases the frequency of high-accuracy reasoning patterns and reduces the prevalence of less effective patterns. Appendix B.3 provides the task-specific analysis.

Across these tasks, the highest-success reasoning patterns consistently become more prevalent after RLVR enhancement, reinforcing the findings in Section 4 and the theoretical account in Section 5. In Algebra and AMC23-19, Figure 4 also shows comparatively stable success rates for individual reasoning patterns, providing additional empirical support for Assumption 5.1.

A.3 Qwen-2.5-32B-Instruct vs. QwQ-32B

We further evaluate the theory on a larger model, QwQ-32B [32], which is based on Qwen-2.5-32B-Instruct [39] and substantially improves reasoning through RLVR. We compare reasoning-pattern shifts between the two models on four tasks: Number Theory and Geometry from MATH [16], one task from AIME 2024 [7], and one task from OlympiadBench [15].

Most reasoning patterns selected by QwQ-32B correspond to the most accurate patterns observed for Qwen-2.5-32B, which is consistent with the theoretical prediction that RLVR shifts probability mass toward higher-success patterns. We do not report QwQ pattern accuracies because API-based testing does not provide the CoT data needed to let the base model continue generation from the same partial reasoning trajectory. Consequently, we cannot measure reasoning-pattern accuracy separately from overall model accuracy in this setting. See the “Accuracy Analysis” paragraph in Appendix B.2 for the corresponding protocol.

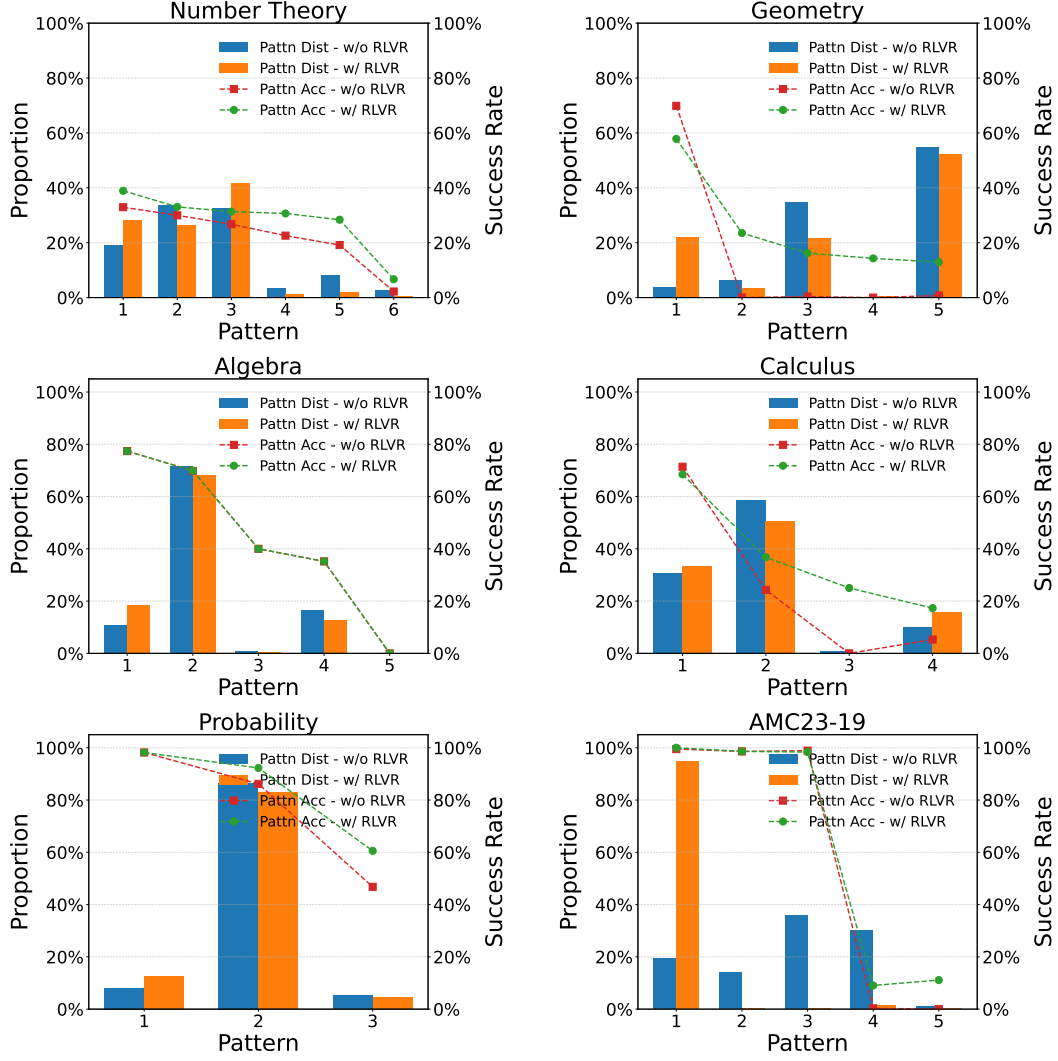
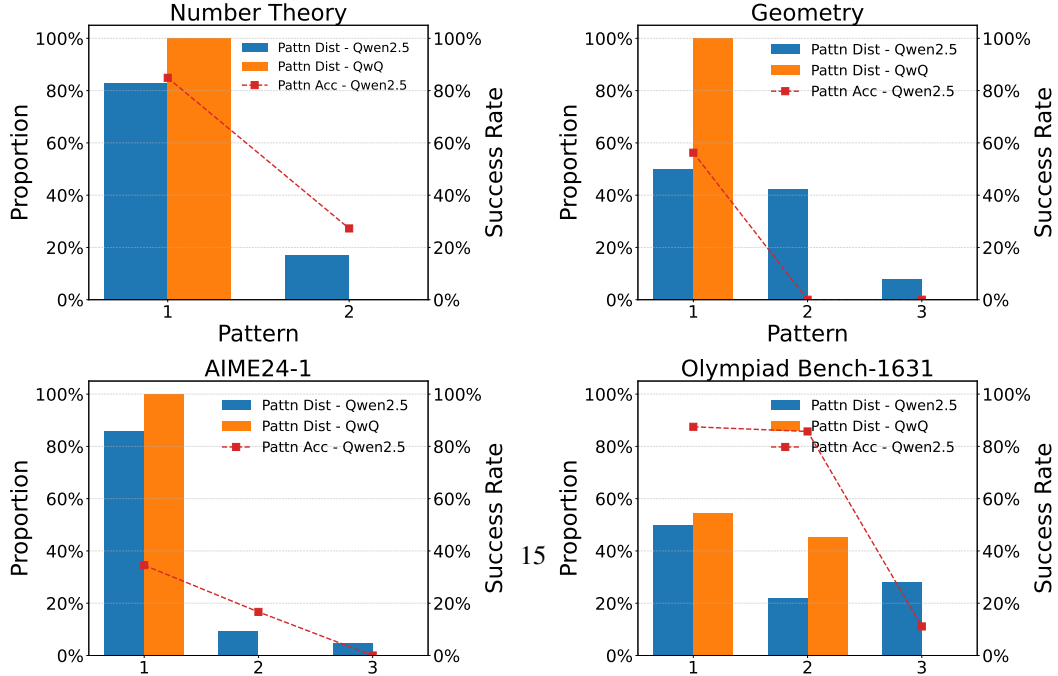


Figure 4: Reasoning-pattern distributions and corresponding success rates for models with and without RLVR enhancement across additional tasks. The bars (Pattern Dist) report each pattern’s proportion among all patterns, and the dots (Pattern Acc) report the corresponding accuracy or success rate. The results align with the main empirical observations and theoretical conclusions.



B Experimental Details

B.1 Experimental Setup

Entire Training Procedures: We use Qwen2.5-3B as the base model and train it on the MATH dataset with both RLVR and RLIF. All experiments use the verl [31] framework, with parameter settings matched to the MATH example provided by verl. For RLIF, the only modification is the reward calculation method. We run three random seeds for each method, yielding three training curves per paradigm.

Token-level Analysis: This experiment uses Qwen2.5-3B and the RL-enhanced models described above. For GSM8K and MATH, we randomly sample 400 questions from each test set; for AIME24, we use the full dataset. We first sample one answer from the base model for each question. We then concatenate this answer with the original question and feed the resulting prompt to the RL-enhanced model. Using the OpenAI library interface, we compute each prompt token’s probability and its rank among all possible next-token probabilities at the corresponding position. Comparing token ranks before and after RL gives the proportion of tokens whose ranking changes.

Reasoning-pattern Level Analysis: We conduct two types of reasoning-pattern experiments. First, using Qwen2.5-3B and its RL-enhanced checkpoints, we evaluate every 20 steps to track how pattern distributions evolve during training. Second, to examine whether the conclusions hold for different base models and larger models, we compare Qwen-2.5-7B-SimpleRL-Zoo, QwQ-32B, and their corresponding base models. For these released models, we can analyze pre- and post-RL distributional differences in model outputs, but not the full within-training trajectory.

B.2 Reasoning Patterns Analysis Pipeline

Model Inference: We randomly sample questions from multiple task types, including Geometry and Algebra, and from different difficulty levels, including Level 4 and Level 5, in datasets such as MATH. For each question, the model samples 1024 answers in the pattern-distribution dynamics experiment. In experiments that compare only the distribution before and after RL, the number of sampled answers ranges from 64 to 1024 depending on the task category. These responses serve as the input for subsequent pattern extraction and classification.

Reasoning Patterns Extraction: We sample a subset of base-model responses and use the GPT-4o [19] API to summarize reasoning-pattern categories from those samples. Appendix D.1 gives the full prompt used for pattern extraction. To make the extraction procedure deterministic, we set the GPT-4o API temperature to 0.

Responses Classification: We then use GPT-4o to classify all responses according to the extracted pattern categories, again with temperature set to 0. Appendix D.2 provides the full classification prompt.

Accuracy Analysis: To isolate the effect of pattern distribution on accuracy, we construct new prompts by concatenating the original question with partial base-model responses that represent specific reasoning patterns. This design fixes the reasoning trajectory up to the final prediction, making it possible to assess whether accuracy changes arise from the distribution over patterns rather than from changes inside the reasoning process itself. Concretely, we remove the sentence containing the final answer from the base-model response, append the preceding sentence to the question, and use this concatenation as a prompt for the RL-enhanced model. The accuracy of the model’s continued prediction is then used as the success rate of the corresponding reasoning pattern.

B.3 Details for Each Task

We now detail the experimental procedures for each task, including task descriptions and examples of reasoning patterns (Table 2). Appendix D.3 provides example task prompts.

Number Theory: This task presents models with problems involving coin distribution across multiple bags. Initially, bags contain equal numbers of coins. After receiving additional coins

Task	Reasoning Patterns Examples	Common Elements / Key Words
Number Theory	Modular Congruence with Coefficient Simplification	Uses modular arithmetic to express divisibility conditions.
	Inequality-Driven Search for Minimal Solution	Often involves substituting back to compute the total coins.
Geometry	Systematic Inequality Application	Explicitly lists and solves each of the three inequalities.
	Verification of Scalene Condition	Adding an extra layer of validation to verifies the third side length.
Algebra	Iterative Floor Division with Leftover Tracking.	Uses floor division to compute new cans per step.
	Recursive Recycling with Aggregated Leftovers.	Combines leftovers with newly produced cans before recycling.
Calculus	Direct Simplification and Principal Value Matching	Assumes inputs fall within the principal range.
	Interval Analysis with Case Splitting	Splits the whole domain into different intervals.
Counting and Probability	Direct Probability Setup and Quadratic Solution	Explicitly calculates combinations for total and favorable outcomes.
	Early Simplification and Cross-Multiplication	Early cross-multiplication to eliminate denominators.
AMC23-Q19	Prime Factorization and Simplification	Counting Digits in Large Numbers.
	Rewriting (8^5) as (2^{15}) (15^5) as $(3^5 \cdot 5^5)$.	Final step of counting digits: $(3 \text{ (from 243)} + 15 \text{ (zeros)} = 18)$.
AIME24-Q1	Direct Equation Setup and Elimination	Straightforward and relies on algebraic manipulation.
	Alternative Equation Formulation and Solving	Expressing variables in terms of others early on.
OlympiadBench-Q1631	Direct Calculation and Empirical Testing	Relying on direct computation and empirical verification.
	Factorization-Based Reasoning	Using algebraic factorization to argue certain terms y_n .

Table 2: Reasoning Patterns Examples for Varying Tasks.

and redistributing them equally, the total must exceed a specified value while maintaining equal distribution. Models must determine the minimum initial coin count per bag. We derived this task from a level 5 MATH dataset problem (id: test/number_theory/1055) [16], creating 32 variants by adjusting parameters like bag count and coin totals as our evaluation dataset. We employ one-shot prompting with a simple, unrelated example to guide answer formatting using "boxed" notation without influencing reasoning approaches. For evaluation, we first choose 20 questions from our evaluation dataset with 4 responses each for reasoning pattern extraction. For each question we sample 64 responses. We then categorize all responses by the extracted reasoning patterns, allowing us to compare changes in the distribution of models' reasoning patterns before and after RL.

Geometry: For the Geometry task, the model is given the lengths of two sides of a scalene triangle and must determine how many integer centimeter values are possible for the third side. The task is derived from a Level 4 MATH geometry problem (id: test/geometry/1046). The evaluation follows the Number Theory pipeline: we construct a synthetic dataset with 32 questions, sample 20 questions and select 4 responses per question from the base model (Qwen2.5-7b-Instruct) for reasoning-pattern extraction, and then compare responses from models with and without RLVR enhancement on all questions in the evaluation set, using 64 responses per question for pattern analysis.

Algebra: The Algebra task is a can-recycling problem. The model starts with a specified number of old cans, and n old cans can be recycled into one new can. The question asks how many cans can ultimately be produced. The difficulty lies in tracking leftovers during each iterative recycling step, since cans that cannot be evenly divided may later be combined and recycled. The task template is derived from a Level 4 MATH algebra problem (id: test/algebra/2768). For this and the following

613 supplementary tasks, we use the same settings as the preceding tasks, including dataset size and
614 number of samples.

615 **Calculus:** The Calculus task presents a single-variable trigonometric equation together with the
616 variable’s domain and asks the model to determine the number of solutions. Although the problem is
617 relatively simple, it probes basic symbolic reasoning for calculus-adjacent tasks. The prototype is a
618 Level 4 MATH pre-calculus problem (id: test/precaculus/1140).

619 **Counting and Probability:** The Counting and Probability task concerns a collection of white and
620 black balls. Two balls are sampled at random; the problem provides the probability that one sampled
621 ball is black and the other is white, along with the number of balls of one color. The model must
622 compute the minimum possible number of balls of the other color. This task is derived from a Level
623 4 MATH counting-and-probability problem (id: test/counting_and_probability/79).

624 **AMC23 Question:** We use question 19 from AMC23 [1]. For reasoning-pattern extraction, we
625 randomly sample 48 responses; for reasoning-pattern analysis, we sample 1024 responses.

626 B.4 Details for Experiments on QwQ-32B and Qwen-2.5-32B

627 We conduct the large-model experiments on four tasks. Because API-based testing limits the number
628 of queries, we randomly select one question per task, sample 64 responses from each of the two
629 models, and directly perform reasoning-pattern extraction and classification on those answers. For
630 the two MATH tasks, we use question 1055 from Number Theory and question 1046 from Geometry.
631 For the remaining tasks, we use problem 1 from AIME24 and question 1631 from Olympiad Bench.
632 Table 2 shows examples of the resulting patterns.

633 C Deferred Proofs

634 This appendix provides the deferred proofs for the main theoretical results: Proposition 5.2, Theo-
635 rems 5.3, Theorems 5.4, and Theorems 5.6.

636 C.1 Proof of Proposition 5.2

637 *Proof of Proposition 5.2.* The proof follows the KL-regularized reward-optimization techniques of
638 [26]. Recall that the RL objective is

$$\phi_{\text{RL}}(\theta) = \mathbb{E}_{\mathbf{x} \sim \mathcal{D}, \mathbf{y} \sim \pi_{\theta}(\mathbf{y} | \mathbf{x})} [r_{\phi}(\mathbf{x}, \mathbf{y})] - \beta \mathbb{D}_{\text{KL}} [\pi_{\theta}(\mathbf{y} | \mathbf{x}) \parallel \pi_{\text{ref}}(\mathbf{y} | \mathbf{x})], \quad (\text{C.1})$$

639 Under the framework introduced in Section 5.1, the policy first samples $\mathbf{r}_i \in \mathcal{R}$ according to $\pi_{\theta}(\mathbf{r}_i | \mathbf{q})$
640 and then generates the final answer $\mathbf{a} \in \mathcal{A}$ according to $\pi_{\theta}(\mathbf{a} | \mathbf{r}_i)$. The objective $\phi_{\text{RL}}(\theta)$ can therefore
641 be written as

$$\phi_{\text{RL}}(\theta) = \mathbb{E}_{\mathbf{y} \sim \pi_{\theta}(\mathbf{y} | \mathbf{q})} \left[\sum_{\mathbf{a} \in \mathcal{A}} \pi_{\text{ref}}(\mathbf{a} | \mathbf{r}) r_{\phi}([\mathbf{r}, \mathbf{a}], \mathbf{q}) \right] - \beta \mathbb{D}_{\text{KL}} [\pi_{\theta}(\mathbf{y} | \mathbf{x}) \parallel \pi_{\text{ref}}(\mathbf{y} | \mathbf{x})], \quad (\text{C.2})$$

642 Let $R(\mathbf{r}) = \sum_{\mathbf{a} \in \mathcal{A}} \pi_{\text{ref}}(\mathbf{a} | \mathbf{r}) r_{\phi}([\mathbf{r}, \mathbf{a}], \mathbf{q})$. Then

$$\begin{aligned} \phi_{\text{RL}}(\theta) &= \mathbb{E}_{\mathbf{r} \sim \pi_{\theta}(\cdot | \mathbf{q}), \mathbf{a} \sim \pi_{\theta}(\cdot | \mathbf{r})} \left[R(\mathbf{r}) - \beta \ln \left(\frac{\pi_{\theta}(\mathbf{r} | \mathbf{q}) \cdot \pi_{\text{ref}}(\mathbf{a} | \mathbf{r})}{\pi_{\text{ref}}(\mathbf{r} | \mathbf{q}) \cdot \pi_{\text{ref}}(\mathbf{a} | \mathbf{r})} \right) \right] \\ &= \mathbb{E}_{\mathbf{r} \sim \pi_{\theta}(\cdot | \mathbf{q})} \left[R(\mathbf{r}) - \beta \ln \left(\frac{\pi_{\theta}(\mathbf{r} | \mathbf{q})}{\pi_{\text{ref}}(\mathbf{r} | \mathbf{q})} \right) \right] \\ &= -\beta \mathbb{E}_{\mathbf{r} \sim \pi_{\theta}(\cdot | \mathbf{q})} \left[\ln \left(\frac{\pi_{\theta}(\mathbf{r} | \mathbf{q})}{\frac{1}{Z} \exp \left(\frac{1}{\beta} R(\mathbf{r}) \right) \pi_{\text{ref}}(\mathbf{r} | \mathbf{q})} \right) - \ln Z \right] \\ &= -\beta \mathbb{D}_{\text{KL}} [\pi^*(\mathbf{r} | \mathbf{q}) \parallel \pi_{\text{ref}}(\mathbf{y} | \mathbf{x})] + \beta \ln Z. \end{aligned}$$

643 The third equality uses Assumption 5.1, and $Z = \sum_{\mathbf{r} \in \mathcal{R}} \exp\left(\frac{1}{\beta} p^*(\mathbf{r})\right) \pi_{\boldsymbol{\theta}_{\text{ref}}}(\mathbf{r}|\mathbf{q})$ is the partition
 644 constant ensuring that

$$\pi^*(\mathbf{r}|\mathbf{q}) = \frac{1}{Z} \exp\left(\frac{1}{\beta} R(\mathbf{r})\right) \pi_{\boldsymbol{\theta}}(\mathbf{r}|\mathbf{q}),$$

645 is a valid probability distribution with $\sum_{\mathbf{r} \in \mathcal{R}} \pi^*(\mathbf{r}|\mathbf{q}) = 1$.

646 Therefore, maximizing the objective in Equation C.2 is equivalent to the following KL minimization
 647 problem:

$$\begin{aligned} & \max_{\pi} \mathbb{E}_{\mathbf{x} \sim \mathcal{D}, \mathbf{y} \sim \pi(\mathbf{y}|\mathbf{x})} [r_{\phi}(\mathbf{x}, \mathbf{y})] - \beta \mathbb{D}_{\text{KL}} [\pi(\mathbf{y} | \mathbf{x}) \parallel \pi_{\text{ref}}(\mathbf{y} | \mathbf{x})] \\ & = \min_{\pi} \beta \mathbb{D}_{\text{KL}} [\pi^*(\mathbf{r}|\mathbf{q}) \parallel \pi_{\text{ref}}(\mathbf{y} | \mathbf{x})] - \beta \ln Z. \end{aligned}$$

648 By non-negativity of KL divergence, the optimal policy for the KL-constrained reward maximization
 649 objective satisfies:

$$\pi_{\text{opt}}(\mathbf{r}|\mathbf{q}) = \frac{1}{Z} \exp\left(\frac{1}{\beta} R(\mathbf{r})\right) \pi_{\boldsymbol{\theta}_{\text{ref}}}(\mathbf{r}|\mathbf{q}) \text{ for all } \mathbf{r} \in \mathcal{R}.$$

650 For RLVR, $R_{\text{RLVR}}(\mathbf{r}) = p^*$. For RLIF, we have

$$\begin{aligned} R'(\mathbf{r}) &= - \left(\frac{1}{|\mathcal{R}|} \sum_{\mathbf{r}_i \in \mathcal{R}} \log(|\mathcal{R}| \cdot \pi_{\boldsymbol{\theta}}(\mathbf{r}_i|\mathbf{q})) + \frac{1}{|\mathcal{A}|} \sum_{\mathbf{a}_j \in \mathcal{A}} \log(|\mathcal{A}| \cdot \pi_{\boldsymbol{\theta}}(\mathbf{a}_j|\mathbf{q}, \mathbf{r})) \right) \\ &= - \frac{1}{|\mathcal{A}|} \sum_{\mathbf{a}_j \in \mathcal{A}} \log(\pi_{\boldsymbol{\theta}}(\mathbf{a}_j|\mathbf{q}, \mathbf{r})) - \underbrace{\left(\frac{1}{|\mathcal{R}|} \sum_{\mathbf{r}_i \in \mathcal{R}} \log(|\mathcal{R}| \cdot \pi_{\boldsymbol{\theta}}(\mathbf{r}_i|\mathbf{q})) + \log|\mathcal{A}| \right)}_c, \end{aligned}$$

651 where the second term is identical for all $\mathbf{r} \in |\mathcal{R}|$. Thus, for RLIF, the optimal policy can be written
 652 as

$$\begin{aligned} \pi_{\text{opt}}(\mathbf{r}|\mathbf{q}) &= \frac{1}{Z} \exp\left(\frac{1}{\beta} R'(\mathbf{r})\right) \pi_{\boldsymbol{\theta}_{\text{ref}}}(\mathbf{r}|\mathbf{q}) \\ &= \frac{\exp\left(\frac{1}{\beta} R'(\mathbf{r})\right) \pi_{\boldsymbol{\theta}_{\text{ref}}}(\mathbf{r}|\mathbf{q})}{\sum_{\mathbf{r}_i \in \mathcal{R}} \exp\left(\frac{1}{\beta} (R'(\mathbf{r}_i))\right) \pi_{\boldsymbol{\theta}_{\text{ref}}}(\mathbf{r}_i|\mathbf{q})} \\ &= \frac{\exp\left(\frac{1}{\beta} (R'(\mathbf{r}) - c)\right) \pi_{\boldsymbol{\theta}_{\text{ref}}}(\mathbf{r}|\mathbf{q})}{\sum_{\mathbf{r}_i \in \mathcal{R}} \exp\left(\frac{1}{\beta} (R'(\mathbf{r}_i) - c)\right) \pi_{\boldsymbol{\theta}_{\text{ref}}}(\mathbf{r}_i|\mathbf{q})} \end{aligned}$$

653 So we can write $R_{\text{RLIF}}(\mathbf{r}) = R'(\mathbf{r}) - c = -\frac{1}{|\mathcal{A}|} \sum_{\mathbf{a}_j \in \mathcal{A}} \log(\pi_{\boldsymbol{\theta}}(\mathbf{a}_j|\mathbf{q}, \mathbf{r}))$. This concludes the proof
 654 of the theorem. $\square$

655 C.2 Dynamics of Tabular Policy

656 Consider the RL objective (Eq 3.1), using a tabular policy (Eq 5.3) with Assumption 5.1 holds, the
 657 gradient dynamics of $\boldsymbol{\theta}$ can be computed as

$$\begin{aligned} \frac{d}{dt} \boldsymbol{\theta}(t) &= \nabla \phi_{\text{RL}}(\boldsymbol{\theta}(t)) \\ &= \nabla \mathbb{E}_{\mathbf{r} \sim \pi_{\boldsymbol{\theta}}(\cdot|\mathbf{q})} [r_{\phi}^{KL}(\mathbf{r}|\mathbf{q})] \\ &= \sum_{\mathbf{r} \in \mathcal{R}} [r_{\phi}^{KL}(\mathbf{r}|\mathbf{q}) \nabla \pi_{\boldsymbol{\theta}(t)}(\mathbf{r}|\mathbf{q})], \end{aligned}$$

Where $\pi_{\theta(t)}(\mathbf{r}|\mathbf{q}) = \text{softmax}(\boldsymbol{\theta}_{:, \mathbf{q}})_{\mathbf{r}}$, $\boldsymbol{\theta}_{:, \mathbf{q}} \in \mathbb{R}^{|\mathcal{V}|}$ is the column of $\boldsymbol{\theta}$ corresponding to $\mathbf{q}$, and $r_{\phi}(\mathbf{r}|\mathbf{q})^{KL}$ refers to the reward with KL divergence, so

$$\begin{aligned} \frac{\partial \boldsymbol{\theta}_{\mathbf{r}_i, \mathbf{q}}(t)}{\partial t} &= \sum_{\mathbf{r}_j \in \mathcal{R}} \left[r_{\phi}^{KL}(\mathbf{r}_j|\mathbf{q}) \left(\nabla \text{softmax}(\boldsymbol{\theta}_{:, \mathbf{q}})_{\mathbf{r}_j} \right)_{\mathbf{r}_i} \right] \\ &= \sum_{\mathbf{r}_j \in \mathcal{R}} \left[r_{\phi}^{KL}(\mathbf{r}_j|\mathbf{q}) \left(-\pi_{\theta(t)}(\mathbf{r}_j|\mathbf{q}) \cdot \pi_{\theta(t)}(\mathbf{r}_i|\mathbf{q}) \right) \right] \\ &\quad + r_{\phi}^{KL}(\mathbf{r}_i|\mathbf{q}) \cdot \pi_{\theta(t)}(\mathbf{r}_i|\mathbf{q}) \end{aligned}$$

where the overall accuracy is defined as $\text{ACC}_{\theta(t)} = \sum_{\mathbf{r} \in \mathcal{R}} \pi_{\theta(t)}(\mathbf{r}|\mathbf{q}) p^*(\mathbf{r})$. For $\pi_{\theta(t)}(\mathbf{r}_i|\mathbf{q})$, we have

$$\frac{d}{dt} \pi_{\theta(t)}(\mathbf{r}_i|\mathbf{q}) = \sum_{\mathbf{r}_j \in \mathcal{R}} \frac{\partial \pi_{\theta(t)}(\mathbf{r}_i|\mathbf{q})}{\partial \boldsymbol{\theta}_{\mathbf{r}_j, \mathbf{q}}(t)} \cdot \frac{\partial \boldsymbol{\theta}_{\mathbf{r}_j, \mathbf{q}}(t)}{\partial t}$$

C.3 Proof of Theorem 5.3

To proof Theorem 5.3, here we first prove that $\text{ACC}_{\theta(t)} \geq \text{ACC}_{\theta(0)}$ for $t \geq 0$ (Eq C.4), then we establish a lower bound for $\pi_{\theta(t)}(\mathbf{r}^*|\mathbf{q})$ (Eq C.5), finally we derive the final bound for t stated in Theorem 5.3.

Proof of Theorem 5.3. Setting $\beta = 0$, Eq C.3 becomes:

$$\frac{\partial \boldsymbol{\theta}_{\mathbf{r}_i, \mathbf{q}}(t)}{\partial t} = \pi_{\theta(t)}(\mathbf{r}_i|\mathbf{q}) \cdot (p^*(\mathbf{r}_i) - \text{ACC}_{\theta(t)}). \quad (\text{C.3})$$

For $\pi_{\theta(t)}(\mathbf{r}_i|\mathbf{q})$, we derive:

$$\begin{aligned} \frac{d}{dt} \pi_{\theta(t)}(\mathbf{r}_i|\mathbf{q}) &= \sum_{\mathbf{r}_j \in \mathcal{R}} \frac{\partial \pi_{\theta(t)}(\mathbf{r}_i|\mathbf{q})}{\partial \boldsymbol{\theta}_{\mathbf{r}_j, \mathbf{q}}(t)} \cdot \frac{\partial \boldsymbol{\theta}_{\mathbf{r}_j, \mathbf{q}}(t)}{\partial t} \\ &= \sum_{\mathbf{r}_j \in \mathcal{R}} (\text{ACC}_{\theta(t)} - p^*(\mathbf{r}_j)) \pi_{\theta(t)}^2(\mathbf{r}_j|\mathbf{q}) \pi_{\theta(t)}(\mathbf{r}_i|\mathbf{q}) \\ &\quad + (p^*(\mathbf{r}_i) - \text{ACC}_{\theta(t)}) \pi_{\theta(t)}^2(\mathbf{r}_i|\mathbf{q}), \end{aligned}$$

and for the accuracy $\text{ACC}_{\theta(t)}$:

$$\begin{aligned} \frac{d}{dt} \text{ACC}_{\theta(t)} &= \sum_{\mathbf{r}_i \in \mathcal{R}} p^*(\mathbf{r}_i) \cdot \frac{d}{dt} \pi_{\theta(t)}(\mathbf{r}_i|\mathbf{q}) \\ &= \sum_{\mathbf{r}_i, \mathbf{r}_j \in \mathcal{R}} p^*(\mathbf{r}_i) \cdot (\text{ACC}_{\theta(t)} - p^*(\mathbf{r}_j)) \pi_{\theta(t)}^2(\mathbf{r}_j|\mathbf{q}) \pi_{\theta(t)}(\mathbf{r}_i|\mathbf{q}) \\ &\quad + \sum_{\mathbf{r}_i \in \mathcal{R}} (p^*(\mathbf{r}_i) - \text{ACC}_{\theta(t)}) \pi_{\theta(t)}^2(\mathbf{r}_i|\mathbf{q}) \\ &= \sum_{\mathbf{r}_j \in \mathcal{R}} \text{ACC}_{\theta(t)} \cdot (\text{ACC}_{\theta(t)} - p^*(\mathbf{r}_j)) \pi_{\theta(t)}^2(\mathbf{r}_j|\mathbf{q}) \\ &\quad + \sum_{\mathbf{r}_i \in \mathcal{R}} (p^*(\mathbf{r}_i) - \text{ACC}_{\theta(t)}) \pi_{\theta(t)}^2(\mathbf{r}_i|\mathbf{q}) p^*(\mathbf{r}_i) \\ &= \sum_{\mathbf{r}_i \in \mathcal{R}} \pi_{\theta(t)}^2(\mathbf{r}_i|\mathbf{q}) (p^*(\mathbf{r}_i) - \text{ACC}_{\theta(t)})^2 \geq 0. \quad (\text{C.4}) \end{aligned}$$

As $\frac{d}{dt} \text{ACC}_{\theta(t)} \geq 0$, so $\text{ACC}_{\theta(t)} > \text{ACC}_{\theta(0)}$ holds for $t \geq 0$, for the optimal reasoning pattern $\mathbf{r}^*$, let $\mathbf{r}'$ the suboptimal reasoning pattern with the second highest success rate, defined as $\mathbf{r}' =$

671 $\arg \max_{\mathbf{r}, \mathbf{r} \neq \mathbf{r}^*} p^*(\mathbf{r})$, we have:

$$\begin{aligned}
\frac{d}{dt} \pi_{\theta(t)}(\mathbf{r}^*|\mathbf{q}) &= \sum_{\mathbf{r}_j \in \mathcal{R}} (\text{ACC}_{\theta(t)} - p^*(\mathbf{r}_j)) \pi_{\theta(t)}^2(\mathbf{r}_j|\mathbf{q}) \pi_{\theta(t)}(\mathbf{r}^*|\mathbf{q}) \\
&\quad + (p^*(\mathbf{r}^*) - \text{ACC}_{\theta(t)}) \pi_{\theta(t)}^2(\mathbf{r}^*|\mathbf{q}) \\
&\geq (\text{ACC}_{\theta(t)} - p^*(\mathbf{r}^*)) \pi_{\theta(t)}^2(\mathbf{r}^*|\mathbf{q}) \pi_{\theta(t)}(\mathbf{r}^*|\mathbf{q}) \\
&\quad + (p^*(\mathbf{r}^*) - \text{ACC}_{\theta(t)}) \pi_{\theta(t)}^2(\mathbf{r}^*|\mathbf{q}) \\
&= (p^*(\mathbf{r}^*) - \text{ACC}_{\theta(t)}) \pi_{\theta(t)}^2(\mathbf{r}^*|\mathbf{q}) (1 - \pi_{\theta(t)}(\mathbf{r}^*|\mathbf{q})) \\
&\geq (p^*(\mathbf{r}^*) - p^*(\mathbf{r}')) \pi_{\theta(t)}^2(\mathbf{r}^*|\mathbf{q}) (1 - \pi_{\theta(t)}(\mathbf{r}^*|\mathbf{q}))^2 \geq 0. \tag{C.5}
\end{aligned}$$

672 As $\frac{d}{dt} \pi_{\theta(t)}(\mathbf{r}^*|\mathbf{q}) \geq 0$, $\pi_{\theta(t)}(\mathbf{r}^*|\mathbf{q}) \geq \pi_{\theta(0)}(\mathbf{r}^*|\mathbf{q})$ holds for $t \geq 0$, define $C =$
673 $(p^*(\mathbf{r}^*) - p^*(\mathbf{r}')) \pi_{\theta(0)}^2(\mathbf{r}^*|\mathbf{q})$:

$$\frac{d}{dt} \pi_{\theta(t)}(\mathbf{r}^*|\mathbf{q}) \geq C (1 - \pi_{\theta(t)}(\mathbf{r}^*|\mathbf{q}))^2.$$

674 This differential inequality yields:

$$\pi_{\theta(t)}(\mathbf{r}^*|\mathbf{q}) > 1 - \frac{1}{Ct + \frac{1}{1 - \pi_{\theta(0)}(\mathbf{r}^*|\mathbf{q})}}.$$

675 For any $\epsilon > 0$, there exists $T_1 = \frac{1}{C} \left(\frac{1}{\epsilon} - \frac{1}{1 - \pi_{\theta(0)}(\mathbf{r}^*|\mathbf{q})} \right) = \mathcal{O}(\frac{1}{\epsilon})$, such that

$$\pi_{\theta(t)}(\mathbf{r}^*|\mathbf{q}) > 1 - \epsilon.$$

676 This concludes the proof of the theorem. $\square$

677 C.4 Proof of Theorem 5.4

678 To proof Theorem 5.4, we first establish an upper bound for $\pi_{\theta(t)}(\mathbf{r}'|\mathbf{q})$ (Eq C.6), then we analyze
679 the ratio $\rho_{\theta(t)}(\mathbf{r}_i) = \frac{\pi_{\theta(t)}(\mathbf{r}_i|\mathbf{q})}{\pi_{\theta(t)}(\mathbf{r}^*|\mathbf{q})}$ (Eq C.7) and prove that for $t > T_0$, $\text{ACC}_{\theta(t)} \geq p^*(\mathbf{r}')$.

680 *Proof of Theorem 5.4.* Let $\mathbf{r}^*$ and $\mathbf{r}'$ be the optimal and second optimal reasoning patterns. For
681 $\pi_{\theta(t)}(\mathbf{r}'|\mathbf{q})$:

$$\begin{aligned}
\frac{d}{dt} \pi_{\theta(t)}(\mathbf{r}'|\mathbf{q}) &= \sum_{\mathbf{r}_j \in \mathcal{R}} (\text{ACC}_{\theta(t)} - p^*(\mathbf{r}_j)) \pi_{\theta(t)}^2(\mathbf{r}_j|\mathbf{q}) \pi_{\theta(t)}(\mathbf{r}'|\mathbf{q}) \\
&\quad + (p^*(\mathbf{r}') - \text{ACC}_{\theta(t)}) \pi_{\theta(t)}^2(\mathbf{r}'|\mathbf{q}) \\
&= \sum_{\mathbf{r}_j \in \mathcal{R}/\{\mathbf{r}^*\}} (\text{ACC}_{\theta(t)} - p^*(\mathbf{r}_j)) \pi_{\theta(t)}^2(\mathbf{r}_j|\mathbf{q}) \pi_{\theta(t)}(\mathbf{r}'|\mathbf{q}) \\
&\quad + (p^*(\mathbf{r}') - \text{ACC}_{\theta(t)}) \pi_{\theta(t)}^2(\mathbf{r}'|\mathbf{q}) (1 - \pi_{\theta(t)}(\mathbf{r}^*|\mathbf{q})) \\
&\leq \sum_{\mathbf{r}_j \in \mathcal{R}/\{\mathbf{r}^*\}} \pi_{\theta(t)}^2(\mathbf{r}_j|\mathbf{q}) \pi_{\theta(t)}(\mathbf{r}'|\mathbf{q}) \\
&\quad + (p^*(\mathbf{r}') - p^*(\mathbf{r}')) \pi_{\theta(t)}(\mathbf{r}^*|\mathbf{q}) \pi_{\theta(t)}^2(\mathbf{r}'|\mathbf{q}) (1 - \pi_{\theta(t)}(\mathbf{r}^*|\mathbf{q})) \\
&\leq \pi_{\theta(t)}(\mathbf{r}'|\mathbf{q}) (1 - \pi_{\theta(t)}(\mathbf{r}^*|\mathbf{q}))^2 + p^*(\mathbf{r}') \pi_{\theta(t)}^2(\mathbf{r}'|\mathbf{q}) (1 - \pi_{\theta(t)}(\mathbf{r}^*|\mathbf{q}))^2 \\
&\leq 2 (1 - \pi_{\theta(t)}(\mathbf{r}^*|\mathbf{q}))^2.
\end{aligned}$$

682 This yields:

$$\pi_{\theta(t)}(\mathbf{r}'|\mathbf{q}) \leq 1 - \frac{1}{2t + \frac{1}{1 - \pi_{\theta(0)}(\mathbf{r}'|\mathbf{q})}}. \tag{C.6}$$

683 Next, define $\rho_{\theta(t)}(\mathbf{r}_i) = \frac{\pi_{\theta(t)}(\mathbf{r}_i|\mathbf{q})}{\pi_{\theta(t)}(\mathbf{r}^*|\mathbf{q})}$, then for all $\mathbf{r}_i \in \mathcal{R}/\{\mathbf{r}^*, \mathbf{r}'\}$, we have

$$\begin{aligned} \frac{d}{dt}\rho_{\theta(t)}(\mathbf{r}_i) &= \frac{\pi_{\theta(t)}(\mathbf{r}^*|\mathbf{q}) \frac{d}{dt}\pi_{\theta(t)}(\mathbf{r}_i|\mathbf{q}) - \pi_{\theta(t)}(\mathbf{r}_i|\mathbf{q}) \frac{d}{dt}\pi_{\theta(t)}(\mathbf{r}^*|\mathbf{q})}{\pi_{\theta(t)}^2(\mathbf{r}^*|\mathbf{q})} \\ &= \frac{\pi_{\theta(t)}(\mathbf{r}_i|\mathbf{q})}{\pi_{\theta(t)}(\mathbf{r}^*|\mathbf{q})} ((p^*(\mathbf{r}_i) - \text{ACC}_{\theta(t)}) \pi_{\theta(t)}(\mathbf{r}_i|\mathbf{q}) - (p^*(\mathbf{r}^*) - \text{ACC}_{\theta(t)}) \pi_{\theta(t)}(\mathbf{r}^*|\mathbf{q})) \\ &\leq -\rho_{\theta(t)}(\mathbf{r}_i) (p^*(\mathbf{r}^*) - \text{ACC}_{\theta(t)}) \pi_{\theta(t)}(\mathbf{r}^*|\mathbf{q}) \leq 0 \end{aligned}$$

684 The last inequality is based on the condition for case 2 and $\text{ACC}_{\theta(t)} > \text{ACC}_{\theta(0)}$, so $p^*(\mathbf{r}_i) <$
 685 $\text{ACC}_{\theta(t)}$ holds for all $\mathbf{r}_i \in \mathcal{R}/\{\mathbf{r}^*\}$, $t \geq 0$. Then as $\frac{d}{dt}\rho_{\theta(t)}(\mathbf{r}_i) \leq 0$, we have:

$$\pi_{\theta(t)}(\mathbf{r}^*|\mathbf{q}) = \frac{1 - \pi_{\theta(t)}(\mathbf{r}'|\mathbf{q})}{\sum_{\mathbf{r}_i \in \mathcal{R}/\{\mathbf{r}'\}} \rho_{\theta(t)}(\mathbf{r}_i)} \geq \frac{1 - \pi_{\theta(t)}(\mathbf{r}'|\mathbf{q})}{\sum_{\mathbf{r}_i \in \mathcal{R}/\{\mathbf{r}'\}} \rho_{\theta(0)}(\mathbf{r}_i)}$$

686 While $\text{ACC}_{\theta(t)} < p^*(\mathbf{r}')$, define $\gamma_{\pi_{\text{ref}}} := \sum_{\mathbf{r} \in \mathcal{R}/\{\mathbf{r}'\}} \frac{\pi_{\theta_{\text{ref}}}(\mathbf{r}|\mathbf{q})}{\pi_{\theta_{\text{ref}}}(\mathbf{r}^*|\mathbf{q})}$ we have:

$$\begin{aligned} \frac{d}{dt}\rho_{\theta(t)}(\mathbf{r}_i) &\leq -\rho_{\theta(t)}(\mathbf{r}_i) (p^*(\mathbf{r}^*) - \text{ACC}_{\theta(t)}) \pi_{\theta(t)}(\mathbf{r}^*|\mathbf{q}) \\ &\leq -\rho_{\theta(t)}(\mathbf{r}_i) (p^*(\mathbf{r}^*) - p^*(\mathbf{r}')) \frac{1 - \pi_{\theta(t)}(\mathbf{r}'|\mathbf{q})}{\gamma_{\pi_{\text{ref}}}} \\ &\leq -\rho_{\theta(t)}(\mathbf{r}_i) \frac{p^*(\mathbf{r}^*) - p^*(\mathbf{r}')}{\gamma_{\pi_{\text{ref}}}} \frac{1}{2t + \frac{1}{1 - \pi_{\theta(0)}(\mathbf{r}'|\mathbf{q})}} \end{aligned}$$

687 Then we have:

$$\frac{d}{dt}\rho_{\theta(t)}(\mathbf{r}_i) \leq -\rho_{\theta(t)}(\mathbf{r}_i) \frac{p^*(\mathbf{r}^*) - p^*(\mathbf{r}')}{\gamma_{\pi_{\text{ref}}}} \frac{1}{2t + \frac{1}{1 - \pi_{\theta(0)}(\mathbf{r}'|\mathbf{q})}}, \quad (\text{C.7})$$

688 this differential inequality leads to:

$$\rho_{\theta(t)}(\mathbf{r}_i) \leq \frac{(1 - \pi_{\theta(0)}(\mathbf{r}'|\mathbf{q}))^{-\Delta/(2\gamma_{\pi_{\text{ref}}})}}{(1/(1 - \pi_{\theta(0)}(\mathbf{r}'|\mathbf{q})) + 2t)^{\Delta/(2\gamma_{\pi_{\text{ref}}})}} \rho_{\theta(0)}(\mathbf{r}_i).$$

689 Where $\Delta = p^*(\mathbf{r}^*) - p^*(\mathbf{r}')$. Recall that:

$$\begin{aligned} \text{ACC}_{\theta(t)} - p^*(\mathbf{r}') &= \sum_{\mathbf{r} \in \mathcal{R}} \pi_{\theta(t)}(\mathbf{r}|\mathbf{q}) p^*(\mathbf{r}) - p^*(\mathbf{r}') \\ &= p^*(\mathbf{r}^*) \left[(p^*(\mathbf{r}^*) - p^*(\mathbf{r}')) + \sum_{\mathbf{r}_i \in \mathcal{R}/\{\mathbf{r}'\}} \rho_{\theta(t)}(\mathbf{r}_i) (p^*(\mathbf{r}_i) - p^*(\mathbf{r}')) \right] \\ &\geq p^*(\mathbf{r}^*) \left[(p^*(\mathbf{r}^*) - p^*(\mathbf{r}')) - \sum_{\mathbf{r}_i \in \mathcal{R}/\{\mathbf{r}'\}} \rho_{\theta(t)}(\mathbf{r}_i) p^*(\mathbf{r}') \right]. \end{aligned}$$

690 Let $C_2 = 1/\Delta$, $C_1 = p^*(\mathbf{r}')/\Delta$, define:

$$T_0 = \frac{1}{2 - 2\pi_{\theta_{\text{ref}}}(\mathbf{r}'|\mathbf{q})} \left((C_1 \cdot \gamma_{\pi_{\text{ref}}})^{2C_2 \cdot \gamma_{\pi_{\text{ref}}}} - 1 \right),$$

691 then:

$$\begin{aligned}
& \sum_{\mathbf{r}_i \in \mathcal{R}/\{\mathbf{r}'\}} \rho_{\boldsymbol{\theta}(T_0)}(\mathbf{r}_i) p^*(\mathbf{r}') \\
& \leq \frac{(1 - \pi_{\boldsymbol{\theta}(0)}(\mathbf{r}'|\mathbf{q}))^{-\Delta/(2\gamma_{\text{ref}})}}{(1/(1 - \pi_{\boldsymbol{\theta}(0)}(\mathbf{r}'|\mathbf{q})) + 2T_0)^{\Delta/(2\gamma_{\text{ref}})}} \sum_{\mathbf{r}_i \in \mathcal{R}/\{\mathbf{r}'\}} \rho_{\boldsymbol{\theta}(0)}(\mathbf{r}_i) p^*(\mathbf{r}') \\
& \leq \frac{(1 - \pi_{\boldsymbol{\theta}(0)}(\mathbf{r}'|\mathbf{q}))^{-\Delta/(2\gamma_{\text{ref}})}}{\left(1/(1 - \pi_{\boldsymbol{\theta}(0)}(\mathbf{r}'|\mathbf{q})) \cdot \left(\frac{\gamma_{\text{ref}} p^*(\mathbf{r}')}{\Delta}\right)^{2\gamma_{\text{ref}}/\Delta}\right)^{\Delta/(2\gamma_{\text{ref}})}} p^*(\mathbf{r}') \sum_{\mathbf{r}_i \in \mathcal{R}/\{\mathbf{r}'\}} \rho_{\boldsymbol{\theta}(0)}(\mathbf{r}_i) \\
& = \frac{1}{\left(\left(\frac{\gamma_{\text{ref}} p^*(\mathbf{r}')}{\Delta}\right)^{2\gamma_{\text{ref}}/\Delta}\right)^{\Delta/(2\gamma_{\text{ref}})}} p^*(\mathbf{r}') \gamma_{\text{ref}} \\
& = \Delta = p^*(\mathbf{r}^*) - p^*(\mathbf{r}').
\end{aligned}$$

692 Finally, for $t > T_0$ we have:

$$\begin{aligned}
\text{ACC}_{\boldsymbol{\theta}(t)} - p^*(\mathbf{r}') & \geq p^*(\mathbf{r}^*) \left[(p^*(\mathbf{r}^*) - p^*(\mathbf{r}')) - \sum_{\mathbf{r}_i \in \mathcal{R}/\{\mathbf{r}'\}} \rho_{\boldsymbol{\theta}(t)}(\mathbf{r}_i) p^*(\mathbf{r}') \right] \\
& \geq p^*(\mathbf{r}^*) [(p^*(\mathbf{r}^*) - p^*(\mathbf{r}')) - (p^*(\mathbf{r}^*) - p^*(\mathbf{r}'))] = 0.
\end{aligned}$$

693 This concludes the proof of the theorem. $\square$

694 C.5 Proof of Theorem 5.6

695 *Proof of Theorem 5.6.* First, in the case $|\mathcal{A}| = 2$ and $\pi_{\boldsymbol{\theta}_{\text{base}}}(\mathbf{r}|q) = \frac{1}{|\mathcal{R}|}$, Assumption 5.5 implies
696 $\frac{1}{|\mathcal{R}|} \sum_{\mathbf{r} \in \mathcal{R}} (p^*(\mathbf{r})) > 0.5$. The derivative of overall accuracy is

$$\begin{aligned}
\frac{d}{dt} \text{ACC}_{\boldsymbol{\theta}(t)} & = \sum_{\mathbf{r}_i \in \mathcal{R}} p^*(\mathbf{r}_i) \cdot \frac{d}{dt} \pi_{\boldsymbol{\theta}(t)}(\mathbf{r}_i|\mathbf{q}) \\
& = \sum_{\mathbf{r}_i \in \mathcal{R}} \pi_{\boldsymbol{\theta}(t)}^2(\mathbf{r}_i|\mathbf{q}) (p^*(\mathbf{r}_i) - \text{ACC}_{\boldsymbol{\theta}(t)}) \left(- \sum_{\mathbf{a} \in \mathcal{A}} \log(\pi_{\boldsymbol{\theta}(t)}(\mathbf{a}|\mathbf{r}_i, q)) - \text{Con}_{\boldsymbol{\theta}(t)} \right),
\end{aligned}$$

697 where $\text{Con}_{\boldsymbol{\theta}(t)} = - \sum_{\mathbf{r}_i \in \mathcal{R}} \pi_{\boldsymbol{\theta}(t)}(\mathbf{r}_i|\mathbf{q}) \sum_{\mathbf{a} \in \mathcal{A}} \log(\pi_{\boldsymbol{\theta}(t)}(\mathbf{a}|\mathbf{r}_i, q))$. Here we assume $\mathcal{A} = \{\mathbf{a}^*, \mathbf{a}'\}$
698 where $\mathbf{a}'$ is the incorrect answer, and $\pi_{\boldsymbol{\theta}(0)}(\mathbf{r}_i|\mathbf{q}) = \frac{1}{|\mathcal{R}|}$ for all $\mathbf{r}_i \in \mathcal{R}$. Then:

$$\begin{aligned}
\left. \frac{d}{dt} \text{ACC}_{\boldsymbol{\theta}(t)} \right|_{t=0} & = \sum_{\mathbf{r}_i \in \mathcal{R}} \frac{1}{|\mathcal{R}|^2} \left(p^*(\mathbf{r}_i) - \frac{1}{|\mathcal{R}|} \sum_{\mathbf{r}_j \in \mathcal{R}} p^*(\mathbf{r}_j) \right) \cdot \\
& \quad \left(- \sum_{\mathbf{a} \in \mathcal{A}} \log(\pi_{\boldsymbol{\theta}(0)}(\mathbf{a}|\mathbf{r}_i, q)) + \frac{1}{|\mathcal{R}|} \sum_{\mathbf{r}_j \in \mathcal{R}} \sum_{\mathbf{a} \in \mathcal{A}} \log(\pi_{\boldsymbol{\theta}(0)}(\mathbf{a}|\mathbf{r}_j, q)) \right) \\
& = \frac{1}{|\mathcal{R}|^2} \sum_{\mathbf{r}_i \in \mathcal{R}} \left(p^*(\mathbf{r}_i) - \frac{1}{|\mathcal{R}|} \sum_{\mathbf{r}_j \in \mathcal{R}} p^*(\mathbf{r}_j) \right) \cdot \left(- \sum_{\mathbf{a} \in \mathcal{A}} \log(\pi_{\boldsymbol{\theta}(0)}(\mathbf{a}|\mathbf{r}_i, q)) \right).
\end{aligned}$$

699 We now take $|\mathcal{R}| \rightarrow +\infty$ with success rates $p^*(\mathbf{r})$ sampled i.i.d. from $U[0, 1]$. By the central
700 limit theorem, for large $|\mathcal{R}|$, the sample mean $\frac{1}{|\mathcal{R}|} \sum_{\mathbf{r} \in \mathcal{R}} (p^*(\mathbf{r}))$ concentrates around its expectation
701 0.5 with deviation of order $\mathcal{O}(1/\sqrt{|\mathcal{R}|})$. Define this deviation as $\delta = \mathcal{O}(1/|\mathcal{R}|^{1/2}) > 0$ so that
702 $\frac{1}{|\mathcal{R}|} \sum_{\mathbf{r} \in \mathcal{R}} (p^*(\mathbf{r})) = 0.5 + \delta$.

703 By the Gibbs conditioning principle ([12], Corollary 7.3.5 and Theorem 7.3.8), as $|\mathcal{R}| \rightarrow \infty$, the
704 empirical distribution of $p^*(\mathbf{r})$ conditioned on the mean $\frac{1}{|\mathcal{R}|} \sum_{\mathbf{r} \in \mathcal{R}} (p^*(\mathbf{r})) = 0.5 + \delta$ converges to

the density $f(x) = 1 + 12\delta(x - 0.5)$, $x \in [0, 1]$. We can therefore approximate the summation over $\mathbf{r}$ by an integral over this limiting distribution, giving

$$\left. \frac{d}{dt} \text{ACC}_{\theta(t)} \right|_{t=0} = \int_0^1 - (1 + 12\delta(x - 0.5)) \cdot (x - 0.5 - \delta) \cdot \log(x(1 - x)) dx = \frac{2}{3}\delta > 0$$

Since $\delta > 0$, the derivative is positive. As $|\mathcal{R}| \rightarrow \infty$, this event holds with probability 1, proving the first statement.

To prove the second statement, we first analyze the RLIF reward function. For an action space of size $|\mathcal{A}| = 2$, the reward for a path $\mathbf{r}$ is proportional to the negative log-likelihood of the policy:

$$R_{\text{RLIF}}(\mathbf{r}) - \frac{1}{|\mathcal{A}|} \sum_{\mathbf{a}_j \in \mathcal{A}} \log(\pi_{\theta}(\mathbf{a}_j | q, \mathbf{r})) = -\frac{1}{2} \log(p^*(\mathbf{r})(1 - p^*(\mathbf{r})))$$

The function $g(p) = p(1 - p)$ is maximized at $p = 0.5$. Therefore, maximizing the reward $R_{\text{RLIF}}(\mathbf{r})$ is equivalent to choosing the path whose success rate $p^*(\mathbf{r})$ is furthest from 0.5.

Let $p_{\min}^* = \min_{\mathbf{r} \in \mathcal{R}} p^*(\mathbf{r})$ and $p_{\max}^* = \max_{\mathbf{r} \in \mathcal{R}} p^*(\mathbf{r})$. The path with the highest reward will be the one with success rate $p_{\min}^*$ if it is further from 0.5 than $p_{\max}^*$ is. This condition is expressed as:

$$0.5 - p_{\min}^* > p_{\max}^* - 0.5 \iff p_{\min}^* + p_{\max}^* < 1$$

Thus, the event $\arg \max_{\mathbf{r}} R_{\text{RLIF}}(\mathbf{r}) = \arg \min_{\mathbf{r}} p^*(\mathbf{r})$ is equivalent to the event $p_{\min}^* + p_{\max}^* < 1$.

We now calculate the probability of this event. The success rates $p^*(\mathbf{r})$ are drawn from the limiting distribution $f(x) = 1 + 12\delta(x - 0.5)$. From the theory of order statistics, the scaled minimum and maximum of $n = |\mathcal{R}|$ samples converge in distribution to independent exponential random variables. Let $Y_n = n \cdot p_{\min}^*$ and $Z_n = n \cdot (1 - p_{\max}^*)$. As $n \rightarrow \infty$:

- Y_n converges to $Y \sim \text{Exp}(\lambda_Y)$, where the rate is the density at the lower bound: $\lambda_Y = f(0) = 1 - 6\delta$.
- Z_n converges to $Z \sim \text{Exp}(\lambda_Z)$, where the rate is the density at the upper bound: $\lambda_Z = f(1) = 1 + 6\delta$.

The condition $p_{\min}^* + p_{\max}^* < 1$ can be rewritten in terms of these asymptotic variables:

$$\frac{Y_n}{n} + \left(1 - \frac{Z_n}{n}\right) < 1 \implies Y_n < Z_n$$

The probability of this event is $P(Y < Z)$. For two independent exponential variables, this is given by:

$$P(Y < Z) = \frac{\lambda_Y}{\lambda_Y + \lambda_Z} = \frac{1 - 6\delta}{(1 - 6\delta) + (1 + 6\delta)} = \frac{1 - 6\delta}{2} = \frac{1}{2} - 3\delta$$

Since $\delta = O(1/|\mathcal{R}|^{1/2})$, in the limit as $|\mathcal{R}| \rightarrow \infty$, $\delta \rightarrow 0$. Therefore, the limiting probability is:

$$\lim_{|\mathcal{R}| \rightarrow \infty} P(\arg \max_{\mathbf{r}} R_{\text{RLIF}}(\mathbf{r}) = \arg \min_{\mathbf{r}} p^*(\mathbf{r})) = \lim_{\delta \rightarrow 0} \left(\frac{1}{2} - 3\delta \right) = \frac{1}{2}$$

This concludes the proof of the second statement.

□

D Full Prompts

D.1 Full Prompt for Reasoning Patterns Extraction

Full Prompt for Reasoning Patterns Extraction

You will analyze multiple AI reasoning processes, showing how different models solve problems.

Analysis Steps:

For each reasoning process, identify:

Key words and recurring phrases

Logical structure of the argument

Problem-solving techniques used

Step-by-step progression

Group similar reasoning processes into exactly 5 patterns based on shared:

Vocabulary patterns (common terms and phrases)

Logical frameworks (how arguments are structured)

Solution approaches (methods used to reach conclusions)

IMPORTANT! You need to categorize general thinking patterns for problem solving, rather than categorizing types of problems themselves. In other words, each type of thinking pattern you give can be applied to any problem, rather than being task-specific. What you need to classify is the way of thinking, not the way of solving problems.

For example, you can classify them into:

1. derivation based on existing knowledge

2. derivation combined with verification

3. independent calculation, suitable for learners starting from the basics,

4. detailed manual derivation, suitable for learners who need to deeply understand each step

5. etc

Or you can classify them into:

1. knowledge call type

2. derivation and verification type

3. etc.

Output Format:

Pattern 1:

Name and description

Defining characteristics

Example processes from input

Common elements

Pattern 2:

Name and description

Defining characteristics

Example processes from input

Common elements

...

There are reasoning processes:

{extracted_responses}

Group similar reasoning processes into exactly 5 patterns. Please provide a classification of reasoning patterns, with each group defined by their common characteristics and examples from these processes.

725

726 D.2 Full Prompt for Responses Classification

Full Prompt for Responses Classification

Please classify each reasoning process into one of the following five patterns. Output the classifications in JSON format.

{patterns_categories}

Here are the reasoning processes to classify:

{extracted_responses}

Please output your classification in this JSON format:

727

Prompt Example for Easy Countdown

<lim_start>system
 You are a helpful assistant. You first thinks about the reasoning process in the mind and then provides the user with the answer.<lim_end>
 <lim_start>
 user Using the numbers [1,2,3,7], create an equation that equals 5. You can only use basic arithmetic addition (+) and subtraction (-), and each number from the list can be used at most once. Show your work in <think> </think> tags. And return the final answer in <answer> </answer> tags, for example, to get 5 from [1,2,3,7], we have <answer> $2 + 3 = 5$ </answer>; to get 13 from [1,2,3,7], we have <answer> $1 + 2 + 3 + 7 = 13$ </answer>.<lim_end>
 <lim_start>assistant
 Let me solve this step by step.
 <think>

729

Prompt Example for Long Arithmetic

<lim_start>system
 You are a helpful assistant. You first thinks about the reasoning process in the mind and then provides the user with the answer.<lim_end>
 <lim_start>user
 Considering the experssion (((((((((16 + 47) + 92) + 84) + 5) + 49) + 82) - 33) - 99) - 69), you need to start from the innermost bracket and work your way out. You need to give the specific calculation process. You have to do a certain amount of calculations on each line, and print each calculation in the format of $a + b = c$ or $a - b = c$, for example " $1 + 2 = 3$ ".<lim_end>
 <lim_start>assistant
 Let me solve this step by step.
 <think>

730

Prompt Example for Number Theory

<lim_start>system
 You are a helpful assistant.<lim_end>
 <lim_start>user
 You have four bags of copper coins. Each bag has the same number of copper coins. One day, you find a bag of 23 coins. You decide to redistribute the number of coins you have so that all five bags you hold have the same number of coins. You successfully manage to redistribute all the coins, and you also note that you have more than 120 coins. What is the smallest number of coins you could have had before finding the bag of 23 coins?
 Please reason step by step, and put your final answer within \boxed{ }.<lim_end>
 <lim_start>assistant

731

Prompt Example for Geometry

<lim_start>system
 You are a helpful assistant.<lim_end>
 <lim_start>user
 Two sides of scalene $\triangle ABC$ measure 4\$ centimeters and 7\$ centimeters. How many different whole centimeter lengths are possible for the third side?
 Please reason step by step, and put your final answer within \boxed{ }.<lim_end>
 <lim_start>assistant

732

Prompt Example for Algebra

<lim_start>system
You are a helpful assistant.<lim_end>
<lim_start>user
Six aluminum cans can be recycled to make a new can. How many new cans can eventually be made from 200 aluminum cans? (Remember that the first new cans that are made can then be recycled into even newer cans!)
Please reason step by step, and put your final answer within $\boxed{}$.<lim_end>
<lim_start>assistant

733

Prompt Example for Calculus

<lim_start>system
You are a helpful assistant.<lim_end>
<lim_start>user
For how many values of x in $[0, 2\pi]$ is $\cos^{-1}(\cos 4x) = \sin^{-1}(\sin x)$?
Please reason step by step, and put your final answer within $\boxed{}$.<lim_end>
<lim_start>assistant

734

Prompt Example for Counting and Probability

<lim_start>system
You are a helpful assistant.<lim_end>
<lim_start>user
7 white balls and k black balls are placed into a bin. Two balls are drawn at random. The probability that one ball is white and the other is black is $\frac{35}{66}$. Find the smallest possible value of k .
Please reason step by step, and put your final answer within $\boxed{}$.<lim_end>
<lim_start>assistant

735

Prompt Example for AMC23-Q19

<lim_start>system
You are a helpful assistant.<lim_end>
<lim_start>user
How many digits are in the base-ten representation of $8^5 \cdot 5^{10} \cdot 15^5$?
Please reason step by step, and put your final answer within $\boxed{}$.<lim_end>
<lim_start>assistant

736

Prompt Example for AIME24-Q1

<lim_start>system
You are a helpful assistant.<lim_end>
<lim_start>user
Every morning Aya goes for a 9 -kilometer-long walk and stops at a coffee shop afterwards. When she walks at a constant speed of s kilometers per hour, the walk takes her 4 hours, including t minutes spent in the coffee shop. When she walks $s+2$ kilometers per hour, the walk takes her 2 hours and 24 minutes, including t minutes spent in the coffee shop. Suppose Aya walks at $s+\frac{1}{2}$ kilometers per hour. Find the number of minutes the walk takes her, including the t minutes spent in the coffee shop.
Please reason step by step, and put your final answer within $\boxed{}$.<lim_end>
<lim_start>assistant

737

Prompt Example for OlympiadBench-Q1631

<lim_start>system

You are a helpful assistant.<lim_end>

<lim_start>user

For a positive integer a , define a sequence of integers $x_1, x_2, \dots$ by letting $x_1 = a$ and $x_{n+1} = 2x_n + 1$ for $n \geq 1$. Let $y_n = 2^{x_n} - 1$. Determine the largest possible k such that, for some positive integer a , the numbers $y_1, \dots, y_k$ are all prime.

Please reason step by step, and put your final answer within $\boxed{}$.<lim_end>

<lim_start>assistant

738